JAIPUR NATIONAL UNIVERSITY

Jagatpura, Jaipur

DEPARTMENT OF COMPUTER AND SYSTEM SCIENCE

BOARD OF STUDIES MEETING

HELD ON

16TH MAY, 2024 2:00 PM

Board of Studies Meeting

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BOARD OF STUDIES IN THE SCHOOL OF ENGINEERING AND TECHNOLOGY

Minutes of the meeting held on 16th May, 2014

The meeting of the Board of Studies in SCHOOL OF COMPUTER & SYSTEM SCIENCES, for the year 2024-25, was held on **16th May 2024** at **2:00 P.M.** in the **GC1- SADTM Campus - School Of Engineering And Technology**, **Jaipur National University** to discuss the launch of BCA in Full Stack Development in association with the industry. Such an association is need of the hour and the products of such programs are best sought after by the industry.

Announcement notification of the meeting with agenda is placed at the end of the Minutes of the Meeting as Annexure – I & II.

The meeting started at 2:00 P.M. and the following members attended the meeting:

S. No.	Name	Designation
1	Prof. Purnima Nag	**Chairman** Sr. Addl. Director, SOET, JNU, Jaipur
2	Prof. Yogesh Chandra Sharma	**Director** Research and Academic Development IQAC & NEP Coordinator
3	Dr. Sunil Gupta	**Member** HoD , School of Computer and System Sciences
4	Ms. Rachana Yadav	**Member** Assistant Professor, SOET , JNU
5	Dr. Shalini Rajawat	**Member** Program Director (Tech), ImaginXP-CLD
6	Mr. Jitendra	**Member** Technical Faculty**,** ImaginXP-CLD
7	Mr. Rohil Chugh	**Member** Technical Faculty**,** ImaginXP-CLD

8	Mr. Umang Tyagi	**Member** Technical Faculty**,** ImaginXP-CLD
9	Ms. Meenakshi Solanki	**External** Senior Full Stack developer SunArc Technologies
10	Ms. Shruti Bijawat	**External** Vertical Data Head, CloudThat
11	Mr. Dharmendra Prajapat	**External** Research Scholar, IIT Roorkee
12	Mr. Manish Bharadwaj	**External** Assistant professor, SKIT

The Chairman of the Board welcomed the members and introduced external BOS Members to the board and further internal members introduced themselves. The members were briefed regarding the Agenda of the BoS meeting, especially the introduction of BCA in Full Stack Development.

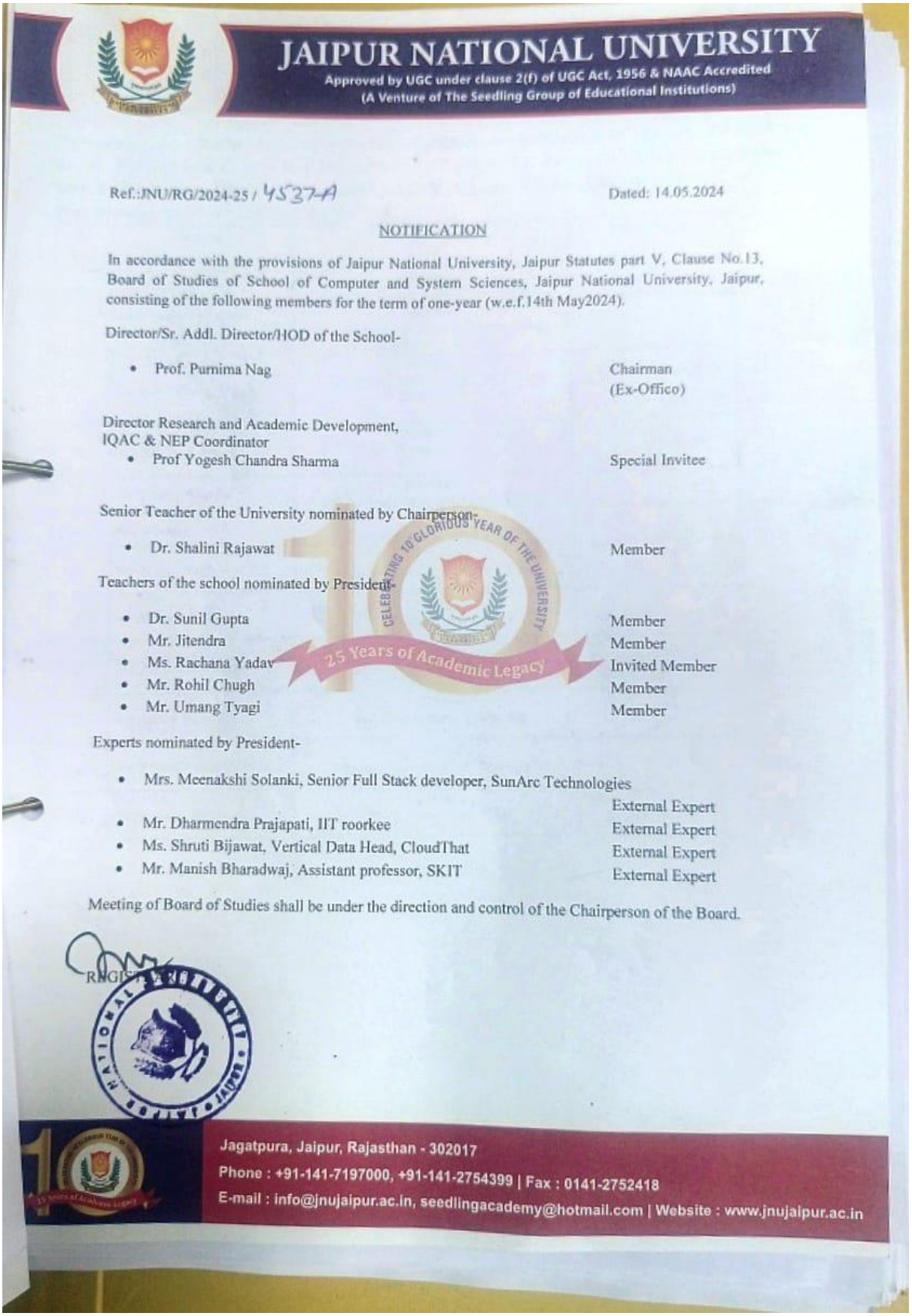
AGENDA NO. 1

To discuss and approve the New Program Structure in BCA in Full Stack Development as per following feedback/suggestions from the session 2024-25:

a) **The program structure, credit framework and courses to be in accordance with the AICTE Guidelines for BCA.**
b) **Number of Credits for the Core Papers to be kept as per Guidelines/Suggestions**
c) **Number of Credits for the Core Papers to be kept 2, 3 or 4. The same will give better in-depth knowledge to the students by increasing the number of courses.**
d) **To include Specialization Course in the program structure and to update the DSE's as per the Discussions during the meeting and under the AICTE guidelines on the same.**
e) **Value Added Courses will be a part of Program Structure with credits that will be the part of the complete credit framework.**

Resolution for Agenda No 1

Agenda No	Resolution No [BOS/SBM/2024/1]	Responsible Person	Target Date	Remarks
1	Unanimously resolved that Proposed **New Program Structure of BCA in Full Stack Development** is accepted and approved with new credits as per AICTE guidelines.	External / Internal members	30th June 2024	Subject to the final approval of Academic Council



JAIPUR NATIONAL UNIVERSITY

Approved by UGC under clause 2(f) of UGC Act, 1956 & NAAC Accredited
(A Venture of The Seedling Group of Educational Institutions)

Ref.:JNU/RG/2024-25/ 4537-A Dated: 14.05.2024

NOTIFICATION

In accordance with the provisions of Jaipur National University, Jaipur Statutes part V, Clause No.13, Board of Studies of School of Computer and System Sciences, Jaipur National University, Jaipur, consisting of the following members for the term of one-year (w.e.f.14th May2024).

Director/Sr. Addl. Director/HOD of the School-

- Prof. Purnima Nag — Chairman (Ex-Offico)

Director Research and Academic Development, IQAC & NEP Coordinator

- Prof Yogesh Chandra Sharma — Special Invitee

Senior Teacher of the University nominated by Chairperson-

- Dr. Shalini Rajawat — Member

Teachers of the school nominated by President-

- Dr. Sunil Gupta — Member
- Mr. Jitendra — Member
- Ms. Rachana Yadav — Invited Member
- Mr. Rohil Chugh — Member
- Mr. Umang Tyagi — Member

Experts nominated by President-

- Mrs. Meenakshi Solanki, Senior Full Stack developer, SunArc Technologies — External Expert
- Mr. Dharmendra Prajapati, IIT roorkee — External Expert
- Ms. Shruti Bijawat, Vertical Data Head, CloudThat — External Expert
- Mr. Manish Bharadwaj, Assistant professor, SKIT — External Expert

Meeting of Board of Studies shall be under the direction and control of the Chairperson of the Board.

REGIS...

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School of Computer and System Sciences

The meeting of the Board of Studies in the School of Computer and System Sciences, for the academic session 2024-25 held on 16th May 2024 at 2:00 pm (In Hybrid Mode) in GC-1 Room (Ground Floor), School of Engineering and Technology, SADTM campus, Jaipur National University, Jaipur.

The following members attended the meeting:

S. No.	Name	Designation	Signature
1	Prof. Purnima Nag	**Chairman** Sr. Addl. Director, SOET, JNU, Jaipur	
2	Prof. Yogesh Chandra Sharma	**Director** Research and Academic Dev IQAC & NEP Coordinator	
3	Dr. Sunil Gupta	**Member** HoD , School of Computer and System Sciences	
4	Ms. Rachana Yadav	**Member** Assistant Professor, SOET , JNU	
5	Dr. Shalini Rajawat	**Member** Program Director (Tech), ImaginXP-CLD	
6	Mr. Jitendra	**Member** Technical Faculty, ImaginXP-CLD	
7	Mr. Rohil Chugh	**Member** Technical Faculty, ImaginXP-CLD	
8	Mr. Umang Tyagi	**Member** Technical Faculty, ImaginXP-CLD	
9	Ms. Meenakshi Solanki	**External** Senior Full Stack developer SunArc Technologies	
10	Ms. Shruti Bijawat	**External** Vertical Data Head, CloudThat	
11	Mr. Dharmendra Prajapat	**External** Research Scholar, IIT Roorkee	
12	Mr. Manish Bharadwaj	**External** Assistant professor, SKIT	

ANNEXURE – I

Ref.: JNU/SOCSS/ BCA(FSD) /2024/01 Date: 14th May, 2024

BOARD OF STUDIES MEETING (BOS)

Notice

The meeting of the Board of Studies in School of Computer & System Sciences, for the **BCA(Full Stack Development)** for the academic session 2024-25 and approval for the same, will be held on 16Th May 2024 at 02:00 pm onwards in GC1, School of Engineering and Technology (SADTM), in hybrid mode, Jaipur National University with the following online bridge details:

Meeting Link: https://meet.google.com/zhe-tsza-hfo

Following agenda will be discussed in the meeting:

1. To discuss and approve the **New Program Structure** in **BCA(Full Stack Development)** as per following feedback/suggestions from the session 2024-25.
2. New **Value-Added Course** from the Session 2024-25
3. Discuss and approve Programme Structure with **Programme Educational Objectives (PEO's)**, Programme Outcomes (PO's) and Course Outcomes (CO's).
4. Identification of **Thrust** area of **Research** in Department of Computer Science and Engineering
5. Review and Discussion on other modifications/suggestions by External members
6. Any other point with the permission of the chair

Dr. Shalini Rajawat
Professor & Course Coordinator

Prof. Purnima Nag
Sr. Addl. Director

ANNEXURE – II

Credit Framework and Program Structure

Program Name: BCA(Full Stack Development)

Syllabus regulation / AY: 2024-25

Course Categories	CREDITS						
	I	II	III	IV	V	VI	Total
Core	11	10	12	15	15	-	63
Discipline-Specific Electives (DSE)	3	4	3	3	3	-	16
Ability-Enhancement Core Courses (AECC)	3	-	3	-	-	-	6
Open Electives (OE)	-	2	-	2	-	-	4
Generic Electives (GE)	-	2	-	3	-	-	5
* Audit Course (AUC)(Value Added Course)	-	-	-	-	-	-	-
Skill Enhancement (SEC)Courses	5	5	5	-	3	-	18
Dissertation/Project	-	-	-	-	-	8	8
Extra-Curricular (EC)			**			-	-
Co-curricular (CC)				**		-	-
Total Credits	22	23	23	23	21	8	120
Total Credits for completion of the programme	22	23	23	23	21	8	**120**

* Qualifying Exam
**Mandatory Qualifying Non-Credited but graded Course

BCA(Full Stack Development)

Program Structure in line with NEP 2020

[L = Lecture, T = Tutorial, P = Practical & C = Credits]

BCA(Full Stack Development) I SEMESTER

Course Code	Course category	Subject Name	Hours per week			Credits	Evaluation		
			L	T	P		IA	ETE	Total
BCAFSCO101T24	CORE	Elementary Mathematics	3	1	0	4	40	60	100
BCAFSDS101T24	DSE1	Fundamentals of Computers and PC Tools	3	0	0	3	40	60	100
BCAFSDS102T24		Information to Cyber Security							
BCAFSDS103T24		Overview of Artificial Intelligence, Ethics and Foundation of Data Analysis							
BCAFSCO102T24	CORE	Programming in C	3	0	0	3	40	60	100
BCAFSSE101T24	SEC	Digital Electronics	3	0	0	3	40	60	100
BCAFSAE101T24	AEC	General English	3	0	0	3	40	60	100
BCAFSCO103P24	CORE	Office Automation Tools Lab	0	0	4	2	40	60	100
BCAFSCO102P24	CORE	Programming in C Lab	0	0	4	2	40	60	100
BCAFSSE101P24	SEC	Digital Electronics Lab	0	0	4	2	40	60	100
	Total		**15**	**1**	**12**	**22**	**320**	**480**	**800**

L: Lecturer, **T**: Tutorial, **IA:** Internal Assessment, **ETE:** End Term Exam

BCA(Full Stack Development)Semester II Theory

Course Code	Course category	Subject Name	Hours per week			Credits	Evaluation		
			L	T	P		IA	ETE	Total
BCAFSDS201T24	DSE2	Discrete Mathematics	3	1	0	4	40	60	100
BCAFSDS202T24		Introduction to information security and internet based attack							
BCAFSDS203T24		Data Analysis and Data Visualisation using Python							
BCAFSCO201T24	CORE	Analysis of Algorithms and Data Structures	3	0	0	3	40	60	100
BCAFSCO202T24	CORE	Web Development	3	0	0	3	40	60	100
BCAFSSE201T24	SEC	Operating System	3	0	0	3	40	60	100
BCAFSVA201T24	VAC	Environmental Science	0	2	0	0	40	60	100
*BCAHVVA202T22	VAC	Human Values and Professional Ethics	0	2	0	0	40	60	100
BCAHPGE201T24	GE2	HTML Programming	2	0	0	2	40	60	100
BCAUXGE202T24		UX Design Fundamentals							
Open Elective	OE	OE1	2	0	0	2	40	40	100
BCAFSCO201P24	CORE	Data Structures and Algorithms Lab	0	0	4	2	40	60	100
BCAFSCO202P24	CORE	Web development Lab with projects	0	0	4	2	40	60	100
BCAFSSE201P24	SEC	Operating System Lab	0	0	4	2	40	60	100
	Total		**16**	**5**	**12**	**23**	**440**	**660**	**1100**

BCA(Full Stack Development) Semester III Theory

Course Code	Course category	Subject Name	Hours per week	Credits	Evaluation

			L	T	P		IA	ETE	Total
BCAFSCO301T24	CORE	Object Orient Programming using C++	3	0	0	3	40	60	100
BCAFSAE302T24	AEC	Computer Networks	3	0	0	3	40	60	100
BCAFSDS301T24	DSE3	Software Engineering	3	0	0	3	40	60	100
BCAFSDS302T24		Network Security							
BCAFSDS303T24		R Programming for data analytics							
BCAFSCO302T24	CORE	Computer Architecture	2	0	0	2	40	60	100
BCAFSCO303T24	CORE	Frontend Development with ReactJS	3	0	0	3	40	60	100
BCAFSSE301T24	SEC	Database Management System	3	0	0	3	40	60	100
	EC*	Extra Curricular Activities							
BCAFSCO301P24	CORE	Object Oriented Programming Lab	0	0	4	2	40	60	100
BCAFSCO303P24	CORE	Frontend Development with ReactJS Lab	0	0	4	2	40	60	100
BCAFSSE301P24	SEC	Database Management Lab	0	0	4	2	40	60	100
		Total	**17**	**0**	**12**	**23**	**360**	**540**	**900**

This is non-credit, Mandatory course, Compulsory to qualify.

L: Lecturer, **T**: Tutorial, **IA:** Internal Assessment, **ETE:** End Term Exam

*Compulsory but not counted towards credit hours

BCA(Full Stack Development) Semester IV Theory

Course Code/	Course category	Subject Name	Hours per week			Credits	Evaluation		
			L	T	P		IA	ETE	Total

BCAFSCO401T24	CORE	Python Programming	3	0	0	3	40	60	100
BCAFSCO402T24	CORE	Backend Development with NodeJS	3	0	0	3	40	60	100
BCAFSCO403T24	CORE	Programming in Java	3	0	0	3	40	60	100
BCAFSDS401T24	DSE4	Cloud Computing	3	0	0	3	40	60	100
BCAFSDS402T24		Mobile Hacking and Malwares							
BCAFSDS403T24		Shell programming with system administration							
BCAFSGE401T24	GE4	E-Commerce Technologies	3	0	0	3	40	60	100
BCAFSGE402T24		Digital Marketing							
Open Elective	OE2	Open Elective-2	2	0	0	2	40	60	100
	CC*	Co Curricular Activity							
BCAFSCO401P24	CORE	Python programming Lab	0	0	4	2	40	60	100
BCAFSCO402P24	CORE	Backend Development with NodeJS Lab	0	0	4	2	40	60	100
BCAFSCO403P24	CORE	Java Programming Lab	0	0	4	2	40	60	100
		Total	**17**	**0**	**12**	**23**	**360**	**540**	**900**

This is non-credit, Mandatory course, Compulsory to qualify.

L: Lecturer, **T**: Tutorial, **IA:** Internal Assessment, **ETE:** End Term Exam

*Compulsory but not counted towards credit hour

BCA(Full Stack Development) Semester V Theory

Course Code	Course category	Subject Name	Hours per week			Credits	Evaluation		
			L	T	P		IA	ETE	Total
BCAFSCO501T23	CORE	Android Programming	3	0	0	3	40	60	100

BCAFSCO502T23	CORE	.Net Programming	3	0	0	3	40	60	100
BCAFSCO503T23	CORE	Data Science Project Management	3	0	0	3	40	60	100
BCAFSSE501T23	SEC	Artificial Intelligence & Machine learning	3	0	0	3	40	60	100
BCAFSVA501T23	VAC	Management Process and Organization behaviour	2	0	0	0	40	60	100
BCAFSDS501T24	DSE5	Data Warehousing and Mining	3	0	0	3	4	60	100
BCAFSDS502T24		Software Reverse Engineering and Cyber forensic							
BCAFSDS503T24		Foundation of Neural Networking and Deep Learning							
BCAFSCO501P24	CORE	Android Programming Lab & Minor Project	0	0	4	2	40	60	100
BCAFSCO502P24	CORE	NET Programming Lab	0	0	4	2	40	60	100
BCAFSCO503P24	CORE	Data Science Project Management.	0	0	4	2	40	60	100
		Total	**17**	**0**	**12**	**21**	**360**	**540**	**900**

L: Lecturer, **T**: Tutorial, **IA:** Internal Assessment, **ETE:** End Term Exam

*Compulsory but not counted towards credit hours

BCA(Full Stack Development) Semester VI

Course Code/Course category	Subject Name	Hours per week	Credits	Evaluation		
				IA	ETE	Total
BCAFSCO601P23	Major Project/Industrial Training	16	8	40	60	100

Total	**16**	**8**	**40**	**60**	**100**

Total Credits = 120

ANNEXURE – III

Programmed Educational Objectives (PEO)

Program: BCA(Full Stack Development)

PEO 1: To expose students to tools and techniques of Computer & System Sciences so that they can comprehend, analyze, design and create innovative computing products and solutions for real life problems.

PEO 2: To develop professional skills in students that prepare them for immediate employment and for lifelong learning in advanced areas of computer science and related fields.

PEO 3: Engage in successful careers in industry, academia, and public service, by applying the acquired knowledge of Science, Mathematics and Engineering, providing technical leadership for their business, profession and community.

Summary sheet for Academic Council

[BOS/SOCSS/ BCA(FSD) /2024/1]: discuss and approve the New Program Structure in BCA(Full Stack Development) as per following feedback/suggestions from the session 2024-25:

Chairman invited suggestions for Introduction of new program. The following changes were requested:

a) The program structure, credit framework and courses to be in accordance with the AICTE Guidelines for BCA(Full Stack Development).

b) Number of Credits for the Core Papers to be kept as per Guidelines/Suggestions

c) Number of Credits for the Core Papers to be kept 2, 3 or 4. The same will give better in-depth knowledge to the students by increasing the number of courses.

d) To include a Specialization Course in the program structure and to update the OE, SEC, AUC, GEGE, CORE, and DSEs as per the Discussions during the meeting and under the AICTE guidelines on the same.

e) Value Added Courses will be a part of Program Structure with credits that will be the part of the complete credit framework.

BCAFSCO101T23	**Elementary Mathematics**	***L T P*** ***3 0 0***	***Credits: 3***	***Marks: 100***

Course Outcomes:

At the completion of the course, a student will be able to:

1. Use elementary algebra, geometry, number concepts, probability and problem solving.
2. Demonstrate familiarity with number theory and statistics.
3. Think mathematically and exhibit confidence in their mathematical ability.
4. Explain why mathematical thinking is valuable in daily life.
5. Represent and statistically analyze data both graphically and numerically.

Detailed Syllabus

UNIT I

Matrices, Types of Matrices, Operations of addition, Scalar Multiplication and Multiplication of Matrices, Determinant of a Square Matrix, Minors and Cofactors, Transpose adjoint and inverse of a matrix, solving system of linear equations in two or three variables using inverse of a matrix

UNIT II

Sets Relation & Functions: Introduction , Definition of Set, Type of Sets, Operations on Sets, Venn diagram, Cartesian Product, Relations, Functions, Types of function, Some elementary functions with their graphs (Exponential logarithmic modulus), Limit & continuity of a function (Simple Problems).

UNIT III

Differentiation: Introduction, Derivative and its meaning, Differentiation of algebraic trigonometric exponential & logarithmic functions, Rules of Differentiation, Differentiation by Substitution, Higher Order Differentiation, Maxima and Minima of Simple Functions

UNIT IV

Integration: Introduction, Integral as Anti-derivative process, Indefinite Integrals, Rules of Integration, Integration by substitution. Definite Integration properties of Definite Integral, Finding areas of Simple Closed Curves

UNIT V

Coordinate Geometry: Introduction, 2D Cartesian Co-ordinate system, Straight line: (Equation & Slope of a line) Circle: Equation of Circle Equation to Tangent, Conic Sections: Focus Eccentricity Directrix, Axis of a conic section, Parabola & Ellipse: (Definitions equations and shape of curve only).

Text Books:

1. Mathematics Vol-2, R. D. Sharma, Dhalpat Raj & Sons.
2. The Elements of Co-ordinate Geometry Part-I, S. L. Loney, Book Palace, New Delhi,

Reference Books:

1. Mathematics for BCA, G. C. Sharma & Madhu Jain, Oscar Publication

BCAFSCO102T23	**Fundamentals of Computers and PC Tools**	***L T P*** ***3 0 0***	***Credits: 3***	***Marks: 100***

Course Outcomes:

At the completion of the course, a student will be able to:

1. Identify the important role of computers and why computers are essential components in business and society along with their various components.
2. Demonstrate the building up of Sequential and combinational logic from basic gates.
3. Apply different categories of programs, system software and applications. Organize and work with files and folders. Utilize the Word Processor, Worksheet and PowerPoint for various applications.
4. Assess the emerging technologies in the area like Big Data, Data Mining and Cloud Computing.
5. Bridge the fundamental concepts of computers with the present level of knowledge of the students.

Detailed Syllabus

Unit I

Computer and its characteristics, application of computers, digital and analog computers, Generation of computers. Storage devices: primary storage devices (RAM, ROM, PROM, EPROM and EEPROM), secondary storage devices (Floppy disk Hard disk optical disk magnetic tapes), Input and output devices (keyboard mouse light pen joystick scanner monitor printers, etc.). Number system and its types

conversion from one base to another and vice versa arithmetic operations r's (r - 1)'s complement methods.

Unit II

Software and its types (System Software, Application Software, Firmware Software), Computer Languages and its types (Machine Language, Assembly Language, High Level Language. Advantages and disadvantages of computer languages) Translators: Compiler Linker Interpreter

Unit III

Overview of Emerging Technologies: Bluetooth, cloud computing, big data ,data mining , mobile computing and embedded systems

Unit IV

Operating system and its functions, types of operating system (Single user, multi-user, multitasking, time sharing, distributed). Fundamental of DOS, internal and external commands, Windows fundamentals: Anatomy of windows, desktop elements, managing files and folders, installing software

Unit V

Word Processor and its features, editing of Text, Find and Replace, Bullets and Numbering, Spell Checker, Grammar Checker, Auto Correct, Auto Complete, Auto Text, Header and footer, tables, mail merge, border and shading, page setup printing. Spread sheet and its features, Entering Information in Worksheet, Editing Cell, Entry Moving and Copying Data, deleting or Inserting Cells Rows and Columns. Custom Numeric Formats Using Formulas and functions, Creating charts. Presentation Software and its uses, steps for creating PowerPoint Presentation, PowerPoint Views Assigning Slide Transitions Using Preset Animations, Hiding Slides, Slide Show, Controlling the Slide Show with a Keyboard and Setting Slide Show Timings.

Text Books:

1. P.K Sinha & Priti Sinha Computer Fundamentals BPB Publications.
2. V. Rajaraman Fundamentals of Computers EEE.
3. Peter Nortan Introduction to Computers Tata Mcgraw Hill.

Reference Books:

1. Alexix Leon Mathewes Leon Fundamentals of Information Technology.

2. Suresh K. Basandra Computer Systems Today Gatgotia Publications.
3. Joyce Coax Joan Preppernau Steve Lambert and Curtis Frye 2007 Microsoft Office System step by step Microsoft Press.
4. R.K. Taxali PC Software for Windows.

BCAFSCO103T23	**Programming in C**	***L T P*** ***3 0 0***	***Credits: 3***	***Marks: 100***

Course Outcomes:

At the completion of the course, a student will be able to:

1. Understand the concept of input and output devices of Computers and how it works and recognize the basic terminology used in computer programming
2. Illustrate concept of compile and debug programs in C language and use different data types for writing the programs.
3. Design programs connecting decision structures, loops and functions.
4. Distinguish between call by value and call by address.
5. Understand the dynamic behavior of memory by the use of pointers.
6. Use different data structures and create / manipulate basic data files and developing applications for real world problems.

Detailed Syllabus

UNIT I

Planning the Computer Program, Debugging, Types of errors, Documentation, Techniques of Problem Solving, Problem solving aspects, Implementation of algorithms, Program verification, Flowcharting, decision table, algorithms and Structured programming concepts; Programming methodologies: Top down and Bottom up programming.

UNIT II

Programming Language, C Standard Library, Introduction to C Programming , Memory Concepts, Decision Making , Secure C Programming , Structured Program Development in C, Algorithms, Pseudocode, Control Structures, if Selection Statement, while Repetition Statement , Assignment Operators, Increment and Decrement Operators. C Program Control: for Repetition Statement, switch Multiple Selection Statement, do...while Repetition Statement, break and continue Statements, Logical Operators.

UNIT III

C Functions, Program Modules in C, Math Library Functions, Functions: Function Definitions, Function Prototypes: A Deeper Look , Function Call Stack and Stack Frames, Passing Arguments By Value and By Reference, Recursion vs. Iteration, C Arrays: Defining Arrays, Passing Arrays to Functions, Sorting Arrays, Searching Arrays and Multidimensional Arrays.

UNIT IV

Structure & Union, C Pointers: Pointer Variable, Definitions and Initialization. Pointer Operators, Passing Arguments to Functions by Reference, size of Operator, Pointer Expressions and Pointer Arithmetic, Relationship between Pointers and Arrays, Pointers to Functions, C Characters and Strings, Character Handling Library, String, Conversion Functions, Standard Input/output Library Functions, String, Manipulation Functions ,C Formatted Input/output

UNIT V

C File Processing: Files and Streams, Creating a Sequential, Access File, Reading Data from a Sequential, Access File , Random, Access Files , Creating a Random, Access File and Writing Data Randomly to a Random Access File, Reading Data from a Random Access File, C Pre-processor.

Text Books:

1. Balagurusamy, Programming in ANSI C, Tata McGraw-Hill Education, 2008
2. Yashavant Kanetkar, Let us C, BPB

Reference Books:

1. P. K. Sinha & Prtti Sinha “Computer ‘5’ Fundamentals” BPB Publications 2007.
2. R.G. Tromey “How to solve it by computer” Prentice Hall 1982.
3. Paul Deital & Harvey Deital “C How to Program” 7 edition Pearson Education 2013.

4. Dr. Anita Goel, Computer Fundamentals, Pearson Education, 2010.

BCAFSSE101T23	Digital Electronics	*L* *3*	*T* *0*	*P* *0*	*Credits: 3*	*Marks: 100*

Course Outcomes:

At the completion of the course, a student will be able to:

1. Recall fundamentals and principles of analog circuits and electronic devices in electrical and electronic engineering. Acquire basic knowledge of physical and electrical conducting properties of semiconductors. Develop the ability to understand the design and working of BJT / FET amplifiers.
2. Employ the codes and number systems converting circuits and compare different types of logic families which are the basic unit of different types of logic gates in the domain of economy, performance, and efficiency.
3. Understand different types of digital electronic circuits using various mapping and logical tools and know the techniques to prepare the most simplified circuit using various mapping and mathematical methods.
4. Analyze, design and implement sequential logic circuits. Assess the nomenclature and technology in the area of memory devices and apply the memory devices in different types of digital circuits for real-world applications.
5. Design different types of with and without memory element digital electronic circuits for a particular operation, within the realm of economic, performance, efficiency, user-friendly and environmental constraints
6. Evaluate frequency response to understand the behavior of Digital electronic circuits. Create and analyze electronic circuits

Detailed Syllabus

Unit I

Introduction to Basic Electronics: Semiconductors, Intrinsic & Extrinsic semi conductors, P Type & N Type semiconductors, PN Junction & Biasing. Semiconductor Diode: Diode, PN junction diode, forward/reverse, current symbol ratings, forward & reverse bias characteristics, Transistor (Introductory concepts): PNP & NPN Transistor, CBCC CE configurations and Transistor as an Amplifier, Introduction to FET MOSFET & construction

Unit II

Data and number representation and Logic Gates: Binary, complement representation, BCD, ASCII, conversion of numbers from one system to the other, 2's complement, representation and binary arithmetic. Logic Gates: AND, OR, NOT, NAND, NOR, Exclusive OR and Exclusive NOR. Implementations of Logic Functions using gates, NAND, NOR implementations, Multi level gate implementations, Multi output gate implementations, Tristate gates

Unit III

Minimization Techniques: Boolean postulates and laws, De Morgan's Theorem, Principle of Duality, Boolean expression, Minimization of Boolean expressions, Minterm, Maxterm, Sum of Products (SOP), Product of Sums (POS), Karnaugh map, Minimization, Don't care conditions, Quine McCluskey method of minimization.

Unit IV

Combinational Circuits: Design procedure, Half adder, Full Adder, Half subtractor, Full subtractor, Parallel binary adder, parallel binary Subtractor, Fast Adder, Carry Look Ahead, adder-Serial Adder/ Subtractor, BCD, adder, Binary Multiplier, Binary Divider Multiplexer/ Demultiplexer, decoder encoder.

Sequential Circuits: Latches, Flipflops, SR, JK, D, T and Master Slave, Characteristic table and equation Application table, Edge triggering - Level Triggering, Introduction to Asynchronous and Synchronous counters and shift registers

Unit V

Memory Devices: Classification of memories, ROM organization, PROM, EPROM, EEPROM, EAPROM, RAM organization, Write operation, Read operation, Memory cycle, Timing wave forms, Memory, decoding memory expansion, Static RAM Cell, Bipolar RAM cell, MOSFET RAM cell, Dynamic RAM cell

Text Books:

1. M. Morris Mano Digital Design 3rd Edition Prentice Hall of India Pvt. Ltd. 2003 Pearson Education (Singapore) Pvt. Ltd. New Delhi 2003.
2. S. Salivahanan and S. Arivazhagan Digital Circuits and Design 3rd Edition. Vikas Publishing House Pvt. Ltd New Delhi 2006.

Reference Books:

1. John F.Wakerly Digital Design Fourth Edition Pearson/PHl 2006.
2. John.M Yarbrough Digital Logic Applications and Design Thomson Learning 2002,
3. Charles H.Roth. Fundamentals of Logic Design Thomson Learning 2003.
4. Donald P.Leach and Albert Paul Malvino Digital Principles and Applications 6thEdition TMH 2003.
5. William H. Gothmann Digital Electronics 2nd Edition PHI 1982.
6. Thomas L. Floyd Digital Fundamentals 8th Edition Pearson Education Inc New Delhi 2003.
7. Donald D. Givone Digital Principles and Design TMH 2003.

BCAFSAE101T23	General English-I	*L* 3	*T* 0	*P* 0	*Credits: 3*	*Marks: 100*

Course Outcomes

At the completion of the course, a student will be able to:

1. Recall various grammatical concepts like tenses, modals, active & passive etc.
2. Differentiate between tenses, modals, prepositions etc.
3. Apply the knowledge of grammar in their day to day conversation.
4. Develop language proficiency by practicing speaking, listening, reading and writing skills.
5. Build a capacity to learn new words to enhance their vocabulary.

Detailed Syllabus

UNIT-I

Grammar-I

1. Sentence Structure
2. Subject & Predicate
3. Tenses

UNIT-II

Grammar-II

1. Prepositions

2. Modals
3. Active & Passive Voice

UNIT-III

Grammar-II

1. Subject-Verb Agreement
2. Punctuations
3. Common Errors

UNIT-IV

1. Paragraph Writing
2. Job Applications (Solicited Unsolicited and Layout)

UNIT-V

1. Reading Comprehension (Reading at various speeds (slow fast very fast) reading different kinds of texts for different purposes (e.g. for relaxation for information for discussion at a later stage etc.); reading between the lines).
2. Effective Listening (Techniques)

Text Books:

1. Quirk & Greenbaum "Advanced English Usage" Pearson Education.
2. www.usingenglish.com- Writing/ Grammar

Reference Books:

1. Banerjee Meera & Mohan Krishna ''Developing Communication Skills" Macmillan Publications 1990.
2. Chaturvedi P.D. "Business Communication" Pearson Publications.

Online Reading/Supporting Material:

1. www.englishcfub.com - Vocabulary Enrichment/ Speaking
2. www.ispeakyouspeak.blogspot.com - Vocabulary Enrichment/ Speaking
3. www.teachertube.com - Writing Technically www.Dictionary.com - Semantic / Grammar.

BCAFSCO103P23	**Programming in C Lab**	***L T P*** ***0 0 4***	***Credits: 2***	***Marks: 100***

Course Outcomes

At the completion of the course, a student will be able to:

1. Identifysituations where computational methods and computers would be useful.
2. Summarize the programming tasks using techniques learned and writepseudo-code.
3. Choose the right data representation formats based on the requirements of the problem.
4. Use comparisons and limitations of the various programming constructs and choose the right one for the task in hand.
5. Implement file Operations in C programming for a given application.

Exercises:

1. Write a C program print Addition / Multiplication of integers.
2. Determining if a number is +ve /, ve / even / odd.
3. Find maximum of 2 numbers from 3 numbers.
4. Construct a program to implement a calculator that reads in integers and operations from the keyboard such as 7 * 3 + 6 = print the answer and exit. Just as on a cheap calculator addition and multiplication have the same precedence and evaluation is strictly left to right. For example 4+5*6 = prints 54 because 4+5=9 is computed before the multiplication.
5. Write a C program to give sum of first n numbers from given n numbers etc.
6. Write a C program to print Integer division.
7. Write a C program for Digit reversing of a number.
8. Write a C program to give factorial of a number.
9. Write a C program to find the sum of individual digits of a positive integer.
10. A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.
11. Write a C program to generate all the prime numbers between 1 and n where n is value supplied by the user's) Write a program which checks a given integer is Fibonacci number or not.
12. Write a C program to find LCM and HCF of a number.
13. Write a C program to generate sine series cosine series etc.
14. Write a C program in C for Pascal Triangle Prime number.
15. Factors of a number. Other problems such as Perfect number GCD of 2 numbers etc. (Write algorithms and draw flowcharts)
16. Write a C program to check whether the number is Palindrome or not.
17. Write a C program in C to check whether the number is leap or not.
18. Write a C program in C to print various diamond patterns.
19. Write a C program in C whether the number is Armstrong or not.
20. Write a program to shift input data by 2 bits left and right.
21. Write a program to use bitwise "&" operator between 2 integer and display the result.

22. Write a program to input 6 numbers and find the biggest and smallest using nested if.
23. Write a program to find the sum of even and odd numbers using switch if else nested if between 1 and 20.
24. Write a program to find the sum of its digits till the result is in single digit
25. Write a program to print the series: x-x3/3!+x5/5!-x7/7...an/n!
    1 + x2/2!-x3/3! + x4/4... xn/n!
26. Write a program to perform Arithmetic operation on an array i.e. Addition Subtraction Multiplication and Division and store the result in another array.
27. Write a program to perform following string operation: with string functions and without string functions.
    Reverse a string, Compare two string, Concatenate two strings
28. Write a program to detect the occurrence of a number in a string.
29. Write a program to accept a string up to 15 character and display the position of a character in a separate line.
30. Write a program to display and count the number of vowels in a string.
31. Write a program to add to pointer addresses of a pointer variable.
32. Write a program to find the factorial of a number using recursion.
33. Write a program to perform different arithmetic operations using pointers
34. Write a program to obtain prime factors of any integer number using functions.
35. Write a program to find the sum of 5 digit number: Without using recursion and with using recursion.
36. Write a program to obtain Fibonacci series by using recursion.
37. Write a program to create, display, modify and append a file (sequential file).
38. Write a program to copy the content of one file to another.
39. Write a program to calculate space in a file (number of blank spaces and not the file size).
40. Write a program to print out the lines from a file that have 50 or more characters in them.
41. Display the initials of first and the middle name separated by "." i.e. Input - Krishna Kumar Singh Output - K .K. Singh
    Now create a main structure that will contain name age salary of an employee.
42. Write a program to call sum ( ) function recursively and perform sum of 1 to 5 numbers.
43. Write a program to find the larger of the two numbers using macro with argument.
44. Write a program to count the number of character, word and lines in a text.
45. Write a program to arrange a list by using any sorting method.

BCAFSCO102P23	Fundamentals of Computers and PC Tools	L T P 0 0 4	Credits: 2	Marks: 100

Course Outcomes

At the completion of the course, a student will be able to:

1. Identify and recall the use of CUI and GUI based operating systems.
2. Summarize the working of various application software's such as MS Word, MS Excel and MS PowerPoint.
3. Apply the various features and functionalities of MS Word, MS Excel and MS PowerPoint.
4. Design and develop various Word files, spreadsheets and PowerPoint presentations.

Exercises for WORD:

1. Create a **telephone directory.**
   - The heading should be 16-point Arial Font in bold
   - The rest of the document should use 10-point font size
   - Other headings should use 10-point Courier New Font.
   - The footer should show the page number as well as the date last updated.
2. Design a time-table form for your college.
   - The first line should mention the name of the college in 16-point Arial Font and should be bold.
   - The second line should give the course name/teacher's name and the department in 14-point Arial.
   - Leave a gap of 12-points.
   - The rest of the document should use 10-point Times New Roman font.
   - The footer should contain your specifications as the designer and date of creation.
3. Create the following one page documents.
   (a) Compose a note inviting friends to a get-together at your house, including a list of things to bring with them.
   (b) Design a certificate in landscape orientation with a border around the document.
4. Create the following document: A newsletter with a headline and 2 columns in portrait orientation, including at least one image surrounded by text.
5. Convert following text to a table, using comma as delimiter

Type the following as shown (do not bold).

Color, Style, Item

Blue, A980, Van

Red, X023, Car

Green, YL724, Truck

Name, Age, Sex

Bob, 23, M

Linda, 46, F

Tom, 29, M

6. Prepare a grocery list having four columns (Serial number, the name of the product, quantity and price) for the month of April, 06.

- Font specifications for Title (Grocery List): 14-point Arial font in bold and italics.
- The headings of the columns should be in 12-point and bold.
- The rest of the document should be in 10-point Times New Roman.

i. Leave a gap of 12-points after the title.

7. XYZ Publications plans to release a new book designed as per your syllabus. Design the first page of the book as per the given specifications.

b. The title of the book should appear in bold using 20-point Arial font.

c. The name of the author and his qualifications should be in the center of the page in 16-point Arial font.

d. At the bottom of the document should be the name of the publisher and address in 16-pomt Times New Roman.

e. The details of the offices of the publisher (only location) should appear in the footer.

8. Create the following one page documents.

a) Design a Garage Sale sign.
b) Make a sign outlining your rules for your bedroom at home, using a numbered list.

9. Enter the following data into a table given on the next page.

Salesperson	Dolls	Trucks	Puzzles
Amit	1327	1423	3193

Shivi	1421	3863	2934
Om	5214	3247	5467
Ananya	2190	1275	1928
Anupama	1201	2528	1203
Maharshi	4098	3079	2067

Add a column Region (values: S, N, N, S, S, S) between the Salesperson and Dolls columns to the given table. Sort your table data by Region and within Region by Salesperson in ascending order.

Exercises for EXCEL

1. Create a student worksheet containing roll numbers, names and total marks. Open a document in Word and insert the excel worksheet using:-

   i) Copy/Paste

   ii) Embedding

   iii) Linking

2. The term wise marks for APS class of 20 students are stored in 3 separate sheets named term 1, term 2 and term 3. Create 4th worksheet that contains student names and their total and average marks for the entire year. Give proper headings using headers. Make the column headings bold and italic. The 4th worksheet should contain college name as the first line. Make it bold, italic and center it.

3. Using a simple pendulum, plot 1-T and 1-T^2 graph.

l	t1	t2	t3	Mean(t)	T=t/20	T^2
70						
80						
90						
100						

1. Consider the following employee worksheet:-

Full Name (First Last)	Grade 1/2/3	Basic Salary	HRA	PF	Gross	Net	(VA) Vehicle Allo...

HRA is calculated as follows:

Grade	HRA % (of Basic)
1	40%
2	35%
3	30%

i) Find max, min and average salary of employees in respective Grade

ii) Count no. of people where VA>HRA

iii) Find out most frequently occurring grade.

iv) Extract records where employee name starts with "A" has HRA>10000

v) Print Grade wise report of all employees with subtotals of net salary and also grand totals. Use subtotal command.

vi) Extract records where Grade is 1 or 2 and salary is between 10000 and 20000 both inclusive.

In a meeting of a marketing department of an organization it has been decided that price of selling an item is fixed at Rs40. It was resolved to increases the sell of more of more items and getting the profit of Rs40,000/.Use Goal Seek of find out how many items you will have to sell to meet your profit figure.

6. To study the variation in volume with pressure for a sample of an air at constant temperature by plotting a graph for P - V and P-I/V. Sample observations are:-

Pressure(P)	**Volume (V)**	**I/V**	**PV**	**P/V**
75	20			
78.9	19			
83.3	18			
88.2	17			

7. Plot the chart for marks obtained by the students (out of 5) vs. frequency (total number of students in class is 50).
8. Create the following worksheets) containing an year wise sale figure of five salesmen in Rs.

Salesman	2002	2003	2004	2005
MOHAN	10000	12000	20000	50000
MITRA	15000	18000	50000	60000
SHIKHA	20000	22000	70000	70000
ROHIT	30000	30000	100000	80000
MANGLA	40000	45000	125000	90000

Apply the following Mathematical & Statistical functions:

i) Calculate the commission for each salesman under the condition :-
   a) If total sales is greater than Rs. 3, 00,000/-, then commission is 10% of total sale made by the salesman.
   b) Otherwise, 4% of total sale.
ii) Calculate the maximum sale made by each salesman.
iii) Calculate the maximum sale made in each year,
iv) Calculate the minimum sale made by each salesman.
v) Calculate the minimum sale made in each year.
vi) Count the no. of sales persons.
vii) Calcu1ate the cube of sales made by Mohan in the year 2002.
viii) Find the difference in sales by salesman Mitra between the year 2002 and 2003. Find the absolute value of difference.
ix) Also calculate the Mode, Std dev, Variance, and Median for the sale made by each salesman.

Calculate the year wise Correlation coefficient between the sales man Mohan and Mitra year wise

. The following table gives a year wise sale figure of five salesmen in Rs.

Salesman	2000	2001	2002	2003
S1	10000	12000	20000	50000
S2	15000	18000	50000	60000
S3	20000	22000	70000	70000
S4	30000	30000	[00000	80000
S5	40000	45000	125000	90000

Calculate total sale year wise.

i) Calculate the net sales made by each salesman
ii) Calculate the commission for each salesman under the condition :-
   a. If total sales is greater than Rs. 4,00,000/-, then commission is 5% of total sale made by the salesman.

   b. Otherwise, 2% of total sale.

iii) Calculate the maximum sale made by each salesman.
iv) Calculate the maximum sale made in each year.
v) Draw a bar graph representing the sale made by each salesman,
vi) Draw a pie graph representing the sale made by salesmen in year 2001.

10. Consider the following worksheet for APS 1st year students:-

S. No.	Name	PH	CH	BY	MT	CS	Total	%	Grade
1									
2									

Grade is calculated as follows:-

If % >=90

Grade A

If % >=80&<90 Grade B

If % >=70&<80 Grade C

If % >=60 &<70 Grade D

i) Calculate Grade using if function
ii) Sort the data according to total marks
iii) Apply filter to display the marks of the students having more than 65% marks.
iv) Draw a pie chart showing % marks scored in each subject by the topper of the class.
v) Draw the doughnut chart of the data as in (iv)
vi) Enter the S. No. of a student and find out the Grade of the student using VLOOKUP.
vii) Extract all records where name
   a) Begins with "A"
   b) Contains "A"
   c) Ends with "A"

11. Create five Power point slides. Each slide should support different format. In these slides explain areas of applications of IT. Make slide transition time as 10 seconds.

12. Create five Power Point slides to give advantages/disadvantages of computer, application of computers and logical structure of computer.

13. Create five Power Point slides detailing the process of internal assessment. It should be a self-running demo.

BCAFSSE101P23	Digital Electronics Lab	*L* 0	*T* 0	*P* 4	*Credits: 2*	*Marks: 100*

Course Outcome:

At the completion of the course, a student will be able to:

1. Recall and learn the basics of logic gates & code conversion.
2. Develop design capability in the binary arithmetic logic circuit
3. Apply knowledge in Combinational Logic Problem formulation and verify their functionalities.
4. Examine design capability in synchronous and asynchronous sequential circuits like flip flops, Shift registers, and counters
5. Evaluate the basic understanding of digital circuits and to verify their operation.

List of Exercises:

1. To study and verify the truth table of logic gates.
2. Design and implementation of 4-Bit Adder and Subtractor using logic gates.
3. Design and implementation of BCD to excess-3 code converter using logic gates,
4. Design and implementation of Binary to gray code converter using logic gates.
5. Design and implementation of 4 bit binary Adder/ Subtractor using IC 7483
6. Design and implementation of 4 bit binary BCD adder using IC 7483
7. Design and implementation of 2 bit Magnitude Comparator using logic gates.
8. Design and implementation of 16 bit odd/even parity checker generator.
9. Design and implementation of multiplexer using logic gates IC74150 and IC74154.
10. Design and implementation of De-multiplexer using logic gates IC74150 and IC74154
11. Design and implementation of encoder using logic gates IC7445 and IC74147
12. Design and implementation of decoder using logic gates 1C7445 and IC74147
13. Construction and verification of 4 bit ripple counter,
14. Design and implementation of 3-bit synchronous up/down counter.
15. Implementation of SISO SIPO PISO and PIPO shift registers using Flip- flops

SEM II

BCAFSCO201T23	**Discrete Mathematics**	***L T P*** ***3 0 0***	***Credits: 3***	***Marks: 100***

Course Outcome:

After completion of this course Student will be able to:

1. Simplify and evaluate basic logic statements including compound statements, implications, inverses, converses, and contra positives using truth tables and the properties of logic.
2. Express a logic sentence in terms of predicates, quantifiers, and logical connectives
3. Apply the operations of sets and use Venn diagrams to solve applied problems; solve problems using the principle of inclusion-exclusion.
4. Determine the domain and range of a discrete or non-discrete function, graph functions, identify one-to-one functions, perform the composition of functions, find and/or graph the inverse of a function, and apply the properties of functions to application problems.
5. Verify that a simple program segment with given initial and final assertions is correct using

the rule of inference for verification of partial correctness and loop invariants.

Detailed Syllabus

Unit I

Introduction: Introduction to Sets, Finite and Infinite Sets, Unaccountably Infinite Sets. Introduction to Functions and relations, Properties of Binary relations, Closure, Partial Ordering Relations

Unit II

Pigeonhole Principle, Permutation and Combinations, Mathematical Induction, Principle of Inclusion and Exclusion

Unit III

Asymptotic Notations, Recurrence Relations: Introduction, Generating Functions, Linear Recurrence Relations with constant coefficients and their solution

Unit IV

Graphs Theory: Basic Terminology of Graphs, Models and Types, Multigraphs, Weighted Graphs, Graph Representation, Graph Isomorphism, Graph Connectivity, Euler and Hamiltonian Paths and Circuits, Planar Graphs, Graph Coloring, Basic Terminology of Trees, Properties of Trees, Spanning Trees

Unit -V

Inference Theory: Introduction, Logical Connectives, Well Formed Formulas, Tautologies, Equivalence

Text Books:

1. C, L Liu and D.P. Mohapatra, Elements of Discrete Mathematics, Third Edition, Tata McGraw Hill, 2008.
2. K. Rosen, Discrete Mathematics and Its Applications, Sixth Edition, Tata McGraw Hi 11, 2007.

Reference Books:

3. T.H. Gormen, C.E. Leiserson, R.L. Rivest, Introduction to Algorithms, Third Edition,Prentice Hall of India,2010.
4. J.P. Trembley, R. Manohar, Discrete Mathematical Structures with Application to Computer Science, First Edition, Tata McGraw Hilt, 2001.
5. David Gries, Fred B. Schneider, A Logical Approach to Discrete Math, Springer; 2010.

Online Reading/Supporting Material:

1. http://ocw.mit.edu/course5/electrical-engineering-and--computer-science/6-042j-mathematics-fpr-computer- science-fal 1-2005/

BCAFSCO202T23	Analysis of Algorithms and Data Structures	*L* 3	*T* 0	*P* 0	*Credits: 3*	*Marks: 100*

Course Outcome:

After completion of this course Student will be able to:

1. Understand basic data structures (such as an array based list, linked list, stack, queue, binary search tree) and algorithms.
2. Acquire the knowledge to analyze, design, apply and use data structures and algorithms to solve engineering problems
3. Evaluate the solutions of problems by implementing them using the advance data structures .
4. Apply modern tool to solve engineering problems using C.
5. Describe an understanding of analysis of algorithms.
6. Synthesize an algorithm or program code or segment that contains iterative constructs and analyze the code segment.

Detailed Syllabus

Unit I

Introduction: Basic Design and Analysis techniques of Algorithms, Correctness of Algorithm

Algorithm Design Techniques: Iterative techniques, Divide and Conquer, Dynamic Programming, Greedy Algorithms, Asymptotic Notations

Unit II

Sorting Techniques: Elementary sorting techniques-Bubble Sort, Insertion Sort, Merge Sort, Advanced **Sorting techniques:** Heap Sort, Quick Sort, Sorting in Linear Time-Bucket Sort, Radix Sort and Count Sort.

Searching Techniques: Linear and Binary search

Complexity Analysis: Medians & Order Statistics.

Unit III

Data Structures: Arrays Single and Multi-dimensional Arrays, Sparse Matrices, Stacks Implementing stack using array and linked list, Prefix, Infix and Postfix expressions, Utility and conversion of these expressions from one to another

Unit IV

Queues: Array and Linked representation of Queue, De-queue, Priority Queues. Linked Lists: Singly, Doubly and Circular Lists, representation of Stack and Queue as Linked Lists.

Unit V

Recursion Developing Recursive Definition of Simple Problems and their implementation; Advantages and Limitations of Recursion. Trees Introduction to Tree as a data structure; Binary Trees, Binary Search Tree, (Creation, and Traversals of Binary Search Trees), Heaps, Red-black trees

Text Books:

1. T.H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein-Introduction to Algorithms, PHI, 3rd Edition 2009.
2. Robert L. Kruse, "Data Structures and Program Design in C++", Pearson.
3. Seymour Lipschutz, Data Structures with C, Mcgraw Hill

Reference Books:

1. Sarabasse & A.V. Gelder Computer Algorithm-Introduction to Design and Analysis, Publisher - Pearson 3rd Edition 1999.
2. Adam Drozdek, "Data Structures and algorithm in C++", Third Edition, Cengage Learning, 2012.
3. SartajSahni, Data Structures, “Algorithms and applications in C++", Second Edition, Universities Press, 2011.
4. Aaron M. Tenenbaum, Moshe J. Augenstein, YedidyahLangsam, "Data Structures Using C and C++:, Second edition, PHI, 2009.
5. D.S Malik, Data Structure using C++,Second edition, Cengage Learning, 2010

BCAFSDS201T23	Web Development	*L* 3	*T* 0	*P* 0	*Credits: 3*	*Marks: 100*

Course Outcome:

After completion of this course Student will be able to:

1. Understand the Basics of HTML , CSS
2. Learn the various tags of HTML
3. Learn to develop the Web page as pet the requirement.
4. Learn the basics of JavaScript

Understand basic data structures (such as an array based list, linked list, stack, queue, binary search tree) and algorithms

Detailed Syllabus

Unit I

Introduction: Overview of HTML, need of HTML, Use of it, HTML Tags: concept of Tag, types of HTML tags, structure of HTML program Text formatting through HTML: Paragraph breaks, line breaks, background and BGcolor attributes Emphasizing material in a web page: Heading styles, drawing lines, text styles. Text styles and other text effects-centering, spacing, controlling font size & color Lists: Using unordered, ordered, definition lists Adding Graphics To HTML Documents: Using Image tag, attributes of Image tag, changing width & height of image

Unit II

TABLES, FRAMES AND LINKING DOCUMENTS Handling Tables: To define header rows & data rows, use of table tag and its attributes. Use of caption tag Linking Documents: Concept of hyperlink, types of hyperlinks, linking to the beginning of document, linking to a particular location in a document, Images as hyperlinks Frames: Introduction To frames, using frames & frameset tags, named frames how to fix the size of a frame, targeting named frames.

Unit III

INTRODUCTION TO HTML5 Features of HTML5: MIME Types, diving in, Detection techniques, Modernizr: An HTML5 Detection Library, Canvas, Canvas Text, Video Video Formats, Local Storage, Web Workers, Offline Web Applications Geolocation, Input Types, Placeholder Text, Form Autofocus, Microdata Elements of HTML5: The Doctype, the Root Element, The Element New Semantic Elements in HTML5, Handling of Unknown Elements by the Browsers Headers, Articles, Dates and Times, Navigation, Footers.

Unit IV

Introduction to CSS: Introducing CSS, font attributes, color and background attributes, text attributes, border attributes, margin related attributes, list attributes Using class and span tag , External Style Sheets

Unit V

INTRODUCTION TO JAVASCRIPT Introduction to scripting: overview of Java Script, advantages, client side java Script, capturing user input, writing JavaScript into HTML Basic JavaScript Techniques: Data types, literals, variables and operators, Java Script arrays, dense array, operators, expressions Java Script Programming Construct: Assignment, data declaration, if, switch, while, for, do while, label, break, Continue, function call, return, with, delete, method invocation.

Text Books:

1. Web Enabled Commercial Application Development Using HTML, DHTML, JavaScript, Perl CGI by Ivan Bayross (bpb publication)
2. HTML5 by Mark Pilgrim O'Reilly publication.

Reference Books:

1. "Learning PHP, MySQL & JavaScript" by Robin Nixon

BCAFSDS201T23	**Operating System**	***Credits: 3***	***Marks: 100***	***L T P*** ***3 0 0***

Course Outcome:

At the completion of the course, a student will be able to:

1. Recall the main components of an OS & describe the important computer system resources functions and the types of Operating Systems.
2. Explain the working of an OS as a resource manager, file system manager, process manager, memory manager and I/O manager and methods used to implement the different parts of OS and understand the factors in OS design.
3. Evaluate the requirement for process synchronization and coordination handled by operating system
4. Categorize memory organization and explain the function of each element of a memory hierarchy and analyze its allocation policies.
5. Conceptualize the components involved in designing a contemporary OS.

Detailed Syllabus

Unit I

Introduction: System Software, Resource Abstraction, OS strategies.

Types of operating systems - Multiprogramming, Batch, Time Sharing, Single user and Multiuser, Process Control & Real Time Systems.

Unit II

Operating System Organization: Factors in operating system design, basic OS functions, implementation consideration; process modes, methods of requesting system services - system Calls and system programs.

Unit III

Process Management: System view of the process and resources, initiating the OS, process address space, process abstraction, resource abstraction, process hierarchy, Thread model, Deadlocks

Scheduling: Scheduling Mechanisms, Strategy selection, non-pre-emptive and pre-emptive strategies.

Unit IV

Memory Management: Mapping address space to memory space, memory allocation strategies, fixed partition, variable partition, paging, virtual memory, Introduction to File Management, IO Management

Unit V

Shell introduction and Shell Scripting: What is shell and various type of shell, Various editors present in Linux, Different modes of operation in vi editor, shell script, Writing and executing the shell script, Shell variable (user defined and system variables), System calls, Using system calls, Pipes and Filters, Decision making in Shell Scripts (If else, switch), Loops in shell, Function, Utility programs (cut, paste, join, tr, uniq utilities), Pattern matching utility (grep)

Text Books:

1. A Silberschatz, P.B. Galvin, G. Gagne, Operating Systems Concepts, 8th Edition, John Wiley Publications 2008.
2. A.S. Tanenbaurn, Modern Operating Systems, 3rd Edition, Pearson Education 2007.

Reference Books:

1. G. Nutt, Operating Systems: A Modern Perspective, 2nd Edition Pearson Education 1997.
2. W, Stallings, Operating Systems, Internals & Design Principles, 5th Edition, Prentice Hall of India. 2008.
3. M. Milenkovic, Operating Systems- Concepts and design, Tata McGraw Hill 1992.

BCAFSVA201T22	Environmental Science	*L* *T* *P* *2* *0* *0*	*Credits: 2*	*Marks: 100*

Course Outcome:

At the completion of the course, a student will be able to:

1. Identify and justify key stakeholders in humanities and social sciences that need to be a part of sustainable solutions.
2. Articulate the interdisciplinary context of environmental issues.

3. Formulate an action plan for sustainable alternatives that integrate science, humanist, and social perspectives.
4. Understand the transnational character of environmental problems and ways of addressing them, including interactions across local to global scales.
5. Access the qualitative and quantitative research methods to gain empirical evidence bearing on evaluation of environmentally sustainable alternatives

Detailed Syllabus

Unit I

Introduction To Environmental Sciences: Natural Resources: Environmental Sciences, Relevance, Significance of Public awareness, Forest resources, Water resources, Mineral resources, Food resources, conflicts over resource sharing, Exploitation, Land use pattern, Environmental impact, fertilizer, Pesticide Problems, case studies.

Unit II

Ecosystem, Biodiversity and Its Conservation: Ecosystem, concept, structure and function producers, consumers and decomposers, Food chain, Food web, Ecological pyramids, Energy flow, Forest, Grassland, desert and aquatic ecosystem

Biodiversity, Definition genetic, species and ecosystem diversity, Values and uses of biodiversity, biodiversity at global, national (India) and local levels Hotspots, threats to biodiversity conservation of biodiversity Insitu & Exsitu

Unit III

Environmental Pollution And Management: Environmental Pollution, Causes, Effects and control measures of Air, Water, Marine, soil, solid waste, Thermal, Nuclear pollution and Disaster Management, Floods, Earth quake, Cyclone and Landslides. Role of individuals in prevention of pollution, pollution case studies

Unit IV

Social Issues, Human Population: Urban issues, Energy water conservation, Environmental Ethics, Global warming, Resettlement and Rehabilitation issues, Environmental legislations, Environmental production Act. 1986 Air, Water, Wildlife and forest conservation Act, Population growth and Explosion, Human rights and Value Education Environmental Health HIV/AIDS - Role of IT in Environment and Human Health, Women and child welfare, Public awareness, Case studies.

Unit V

Visit to a local area / local polluted site / local simple ecosystem Report submission.

Books Recommended:

1. Kumarasamy, k., A Alagappa Moses and M.Vasanthy, 2004, Environmental studies, Bharathisadan University Publication, Trichy.
2. Kalavathy, s. (ed.) 2004, Environmental studies, Bishop Heber College Publication, Trichy.

Reference Books:

1. Rajamannar, 2004, environmental studies, Ever College Publication, Trichy.

BCAHVVA202T22	Human Values and Professional Ethics	*L* *3* *T* *0* *P* *0*	*Credits: 0*	*Marks: 100*

Course Outcome:

After successful completion of the course, students will be able to:

1. Understand the significance of value inputs in a classroom and start applying them in their life and profession

2. Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body, Intention and Competence of an individual, etc.

3. Understand the role of a human being in ensuring harmony in society and nature.

4. Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Unit I

Human Values: Morals, Values, and Ethics, Integrity, Trustworthiness, Work Ethics, Service-Learning, Living Peacefully, Honesty, Courage, Caring, Sharing, Value Time, Co-operation, Commitment, Civic Virtue, Respect for others, Self-confidence, Empathy, Spirituality, Character.

Unit II

Principles for Harmony: Truthfulness, Customs and Traditions, Human Dignity, Value Education, Human Rights, Fundamental Duties, Aspirations, and Harmony (I, We & Nature), Emotional Intelligence, Gender Bias, Mayer Model, Emotional Competencies, Conscientiousness

Unit III

Engineering Ethics and Social Experimentation: History of Ethics, Need of Engineering Ethics, Senses of Engineering Ethics, Profession, and Professionalism, Self Interest, Moral Autonomy, Utilitarianism, Uses of Ethical Theories, Virtue Theory, Types of Inquiry, Deontology, Kohlberg's Theory, Heinz's Dilemma, Gilligan's Argument, Learning from the Past, Comparison with Standard Experiments, Consultants and Leaders, Engineers as Managers, Role of Codes, Balanced Outlook on Law, Codes and Experimental Nature of Engineering.

Unit IV

Engineers' Responsibilities towards Safety and Risk : The concept of Safety, Safety and Risk, Types of Risks, Voluntary v/s Involuntary Risk, Consequences, Risk Assessment, Liability, Accountability, Reversible Effects, Delayed v/s Immediate Risk, Threshold Levels of Risk

Engineers' Duties and Rights: Professional Duties, Collegiality, Techniques for Achieving Collegiality, Senses of Loyalty, Consensus and Controversy, Confidential and Proprietary Information, Professional and Individual Rights, Conflict of Interest, Ethical egoism, Collective Bargaining, Confidentiality, Gifts and Bribes, Occupational Crimes, Problem-solving, Industrial Espionage, Price Fixing, Whistle Blowing

Unit V

Global Issues: Globalization and MNCs, Business Ethics, Cross Culture Issues, Media Ethics, Endangering Lives, Environmental Ethics, Bio-Ethics, Computer Ethics, War Ethics, Research Ethics, Intellectual Property Rights

Text Books:

1. Professional Ethics by R. Subramaniam – Oxford Publications, New Delhi.
2. Engineering Ethics by Harris, Pritchard, and Rabins, Cengage Learning, New Delhi.
3. Human Values And Professional Ethics by Jayshree Suresh and B. S. Raghavan, S.Chand Publications
4. Ethics in Engineering by Mike W. Martin and Roland Schinzinger – Tata McGraw-Hill – 2003.

Reference Books:

1. Professional Ethics and Morals by Prof.A.R.Aryasri, DharanikotaSuyodhana – Maruthi Publications.
2. Human Values & Professional Ethics by S. B. Gogate, Vikas Publishing House Pvt. Ltd., Noida.
3. Professional Ethics and Human Values by A. Alavudeen, R.Kalil Rahman, and M. Jayakumaran – University Science Press.
4. Engineering Ethics & Human Values by M.Govindarajan, S.Natarajan, and V.S.SenthilKumar-PHI Learning Pvt. Ltd – 2009.

5. Professional Ethics and Human Values by Prof.D.R.Kiran-Tata McGraw-Hill – 2013

BCAFSCO202P23	**Data Structures and Algorithms Lab**	***L***	***T***	***P***	***Credits: 2***	***Marks: 100***
		0	***0***	***4***		

Course Outcome:

At the completion of the course, a student will be able to:

- Understand the implementation of recursive functions and strings.
- Introduce the implementation of linked list and the various operations.
- Implement stack and queue using dynamic memory allocation.
- Introduce the Binary Search Tree implementation using C.
- Learn to implement various sorting and searching algorithms.

LIST OF PROGRAMS

1. Given with two numbers 36 and 60, write recursive function using C to find GCD of two numbers.
2. Write a recursive function using C program to display a series such that the sum of two consecutive numbers equals the next number in series. Let the first two numbers be 0 and 1.
3. Use a recursive function for the towers of Hanoi with three discs.
4. Two strings “Welcome” and “World” are provided. Write a program using pointers to find the length of a string and to concatenate the two strings.
5. String1 is stored with “Greater”. Write a program using pointers to copy string1 to string2 and to extract “Great” from string2.
6. Consider an array in following order:

25, 33, 53, 65, 83, 87 and 92. Write a C program to insert the value 88 and remove 65.

7. Write a C Program using dynamic variables and pointers, to construct a singly linked list consisting of the following information n each node: student id (integer), student name (character string) and semester (integer). The operations to be supported are:

a. The insertion operation

   i. At the front of a list

   ii. At the back of the list

   iii. At any position in the list

b. Deleting a node based on student id. If the specified node is not present in the list an error message should be displayed. Both the options should be demonstrated.

c. Searching a node based on student id and updating the information content. If the specified node is not present in the list an error message should be displayed. Both situations should be displayed.

d. Displaying all the nodes in the list.

8. The heights of ten students were marked as 163cm, 171 cm, 158 cm, 167cm,

175cm, 160cm, 173 cm, 149 cm, 180cm and 154cm. Find the difference while sorting the given heights in ascending or descending order using

- Insertion sort
- Selection Sort
- Merge Sort

9. Write a C program to implement the following operations in stacks:

- Push
- Pop
- Display

10. Write a C program to implement the following operations in queue:

- Insert
- Delete
- Display

11. Create a binary search tree and traversing it using in order, pre - order and post order.

12. Create a scenario based on real time domain.

BCAFSDS201P23	Web development Lab with projects	*L* *0*	*T* *0*	*P* *4*	*Credits: 2*	*Marks: 100*

Course Outcome:

At the completion of the course, a student will be able to:

1. Students learn about the concept of web application creation using HTML.
2. Different formatting options used while creating the web pages.
3. Build a responsive web site using HTML5 and CSS3.
4. Understand the role of HTML and CSS in effective web development along with its uses.
5. Develop an effective web application using HTML and CSS as per the projects.

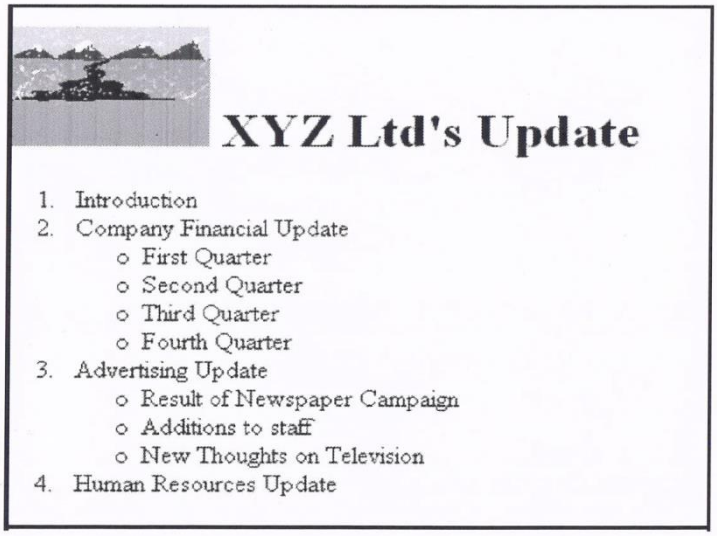
Exercises:

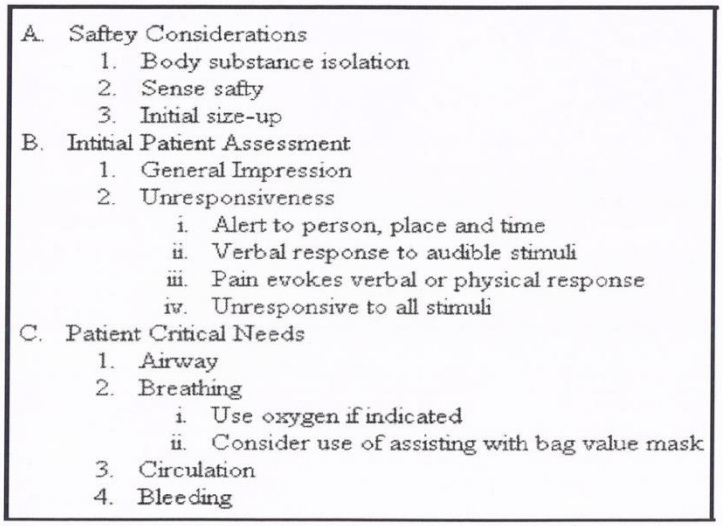
Q.1 Create an HTML document with the following formatting options:

1. Bold
2. Italics
3. Underline
4. Font (Type, Size and Color)
5. Pre tag

Q.2 Create an HTML document which consists of:

1. Ordered List
2. Unordered List
3. Nested List
4. Image


XYZ Ltd's Update
1. Introduction
2. Company Financial Update
o First Quarter
o Second Quarter
o Third Quarter
o Fourth Quarter
3. Advertising Update
o Result of Newspaper Campaign
o Additions to staff
o New Thoughts on Television
4. Human Resources Update


A. Saftey Considerations
1. Body substance isolation
2. Sense safty
3. Initial size-up
B. Intitial Patient Assessment
1. General Impression
2. Unresponsiveness
i. Alert to person, place and time
ii. Verbal response to audible stimuli
iii. Pain evokes verbal or physical response
iv. Unresponsive to all stimuli
C. Patient Critical Needs
1. Airway
2. Breathing
i. Use oxygen if indicated
ii. Consider use of assisting with bag value mask
3. Circulation
4. Bleeding

Q.3 Create an HTML document which implements Internal linking as well as External linking.

Q.4 Create a table using HTML which consists of columns for Roll No., Student's name and grade.

Result		
Roll No.	Name	Grade

Q.5 Create a Table with the following view:

			Place an image here	

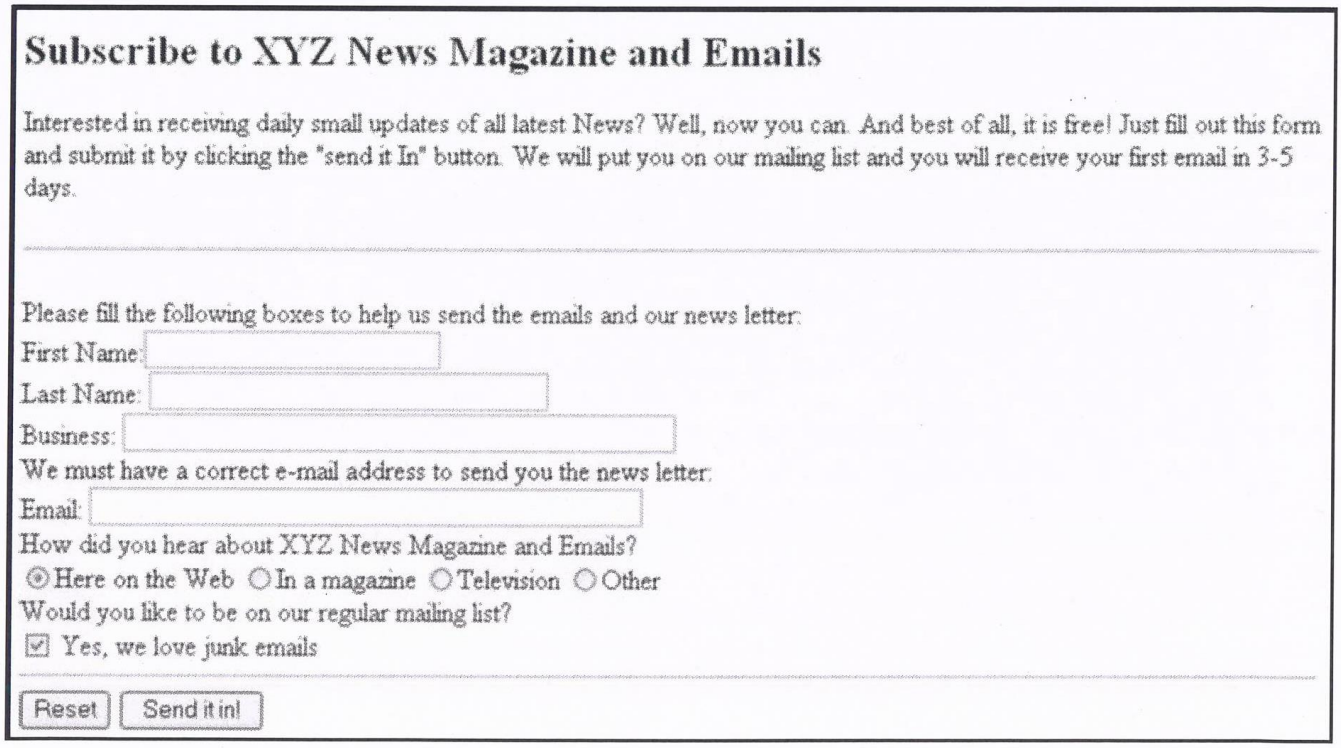
Q.6 Create a form using HTML which has the following types of controls:

I. Text Box

II. Option/radio buttons

III. Check boxes

IV Reset and Submit buttons



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We must have a correct e-mail address to send you the news letter:

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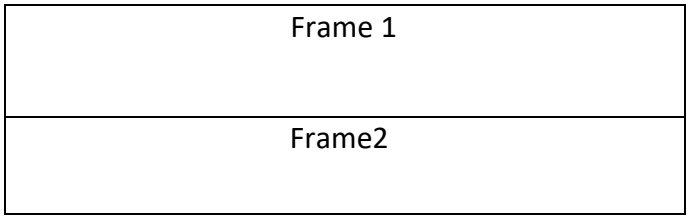
◉ Here on the Web ○ In a magazine ○ Television ○ Other

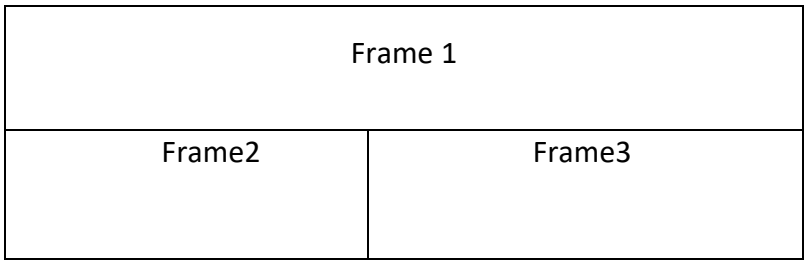
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Reset Send it in!

Q.7 Create HTML documents (having multiple frames) in the following three formats:





List of projects using HTML and CSS(Only for web page and UI design):

1) Interactive response system like any CRM of MNC
2) Railway project administration system
3) Worker loan management system
4) A catalog management system
5) School management system
6) Hospital management system
7) Cyber Café ID system daily Report
8) Internet and Data report login system
9) Mobile dealership management system
10) Garment shop barcode with billing system
11) Coffee shop management system
12) School library system
13) College organization system
14) Travel Reservation system
15) Radio record system

BCAFSSE201P23	Operating System Lab	L T P 0 0 4	Credits: 2	Marks: 100

Course Outcomes:

At the completion of the course, a student will be able to:

1. Remember the functions, structures and history of operating systems.
2. Understand of design issues associated with operating systems
3. Apply concepts including scheduling, synchronization and deadlocks.
4. Distinguish multithreading , Multitasking & Multiprogramming and also able to explain the concept of memory management including virtual Evaluate the requirement for process synchronization and coordination handled by operating system
5. Categorize memory organization and explain the function of each element of a memory hierarchy and analyze its allocation policies.
6. Conceptualize the components involved in designing a contemporary OS.

Experiment List

1. Simulate the following CPU scheduling algorithms.

a) FCFS b) SJF c) Round Robin d) Priority.

2. Write a C program to simulate producer-consumer problem using Semaphores

3. Write a C program to simulate the concept of Dining-philosophers problem.

4. Write a C program to simulate the following contiguous memory allocation Techniques

a) Worst fit b) Best fit c) First fit.

5. Simulate all page replacement algorithms

a)FIFO b) LRU c) OPTIMAL

6. Simulate all File Organization Techniques

a) Single level directory b) Two level directory

7. Simulate all file allocation strategies

a) Sequential b) Indexed c) Linked.

8. Simulate Bankers Algorithm for Dead Lock Avoidance.

9. Simulate Bankers Algorithm for Dead Lock Prevention.

10. Write a C program to simulate disk scheduling algorithms.

a) FCFS b) SCAN c) C-SCAN

SEM III

BCAFSCO301T23	**Object Orient Programming using C++**	***L***	***T***	***P***	***Credits: 3***	***Marks: 100***
		3	***0***	***0***		

Course Outcomes:

At the completion of the course, a student will be able to:

1. Have profound knowledge of object oriented programming.
2. Demonstrate the difference between the solutions offered by traditional imperative
3. Illustrate problem solving methods and object-oriented methods.
4. Explain the class inheritance, data encapsulation, polymorphism as fundamental building blocks to generate reusable code.

5. Understand and implement error handling and file handling routines.

Detailed Syllabus

Unit I

Different paradigms for problem solving, differences between OOP and Procedure oriented programming, Abstraction, Overview of OOP principles, Encapsulation, Inheritance and Polymorphism.

C++ Basics: Structure of a C++ program, Data types, Declaration of variables, Expressions, Operators, Operator Precedence, Evaluation of expressions, Type conversions, Pointers, Arrays, Pointers and Arrays, Strings.

Unit II

Flow control statement- if, switch, while, for, do, break, continue, got statements. Functions-Scope of variables, Parameter passing, Default arguments, inline functions, Recursive functions, Pointers to functions, Dynamic memory allocation and de-allocation operators-new and delete, Preprocessor directives

C++ Classes And Data Abstraction: Class definition, Class structure, Class objects, Class scope, this pointer, Friends to a class, Static class members, Constant member functions, Constructors and Destructors

Unit III

Polymorphism: Function overloading, Operator overloading, Generic programming, necessity of templates, Function templates and class templates.

Inheritance: Defining a class hierarchy, Different forms of inheritance. Defining the Base and Derived classes, Access to the base class members, Base and Derived class construction. Destructors, Virtual base class.

Unit IV

Virtual Functions and Polymorphism: Static and Dynamic bindings, Base and Derived class virtual functions, Dynamic binding through virtual functions, Virtual function call mechanism, Pure virtual functions.

Unit V

C++ I/O: I/O using C functions, Stream classes' hierarchy, Stream I/O, File streams and String streams, Overloading « and » operators, Error handling during file operations, Formatted I/O.

Exception Handling: Benefits of exception handling, throwing an exception, the try block, catching an exception

Text Books:

1. Problem solving with C++, The OOP, 4th Edition, Walter Savitch, Pearson Education.
2. C++, The Complete Reference, 4th Edition, Herbert Schildt, TMH.

Reference Books:

1. C++ Primer, 3rd Edition, S.B.Lippman and J.Lajoie, Pearson Education.
2. The C++ Programming Language, 3rd Edition, B.Stroutstrup, Pearson Education.
3. OOP in C++, 3rd Edition, T.Gaddis, J.Walters and G.Muganda, Wiley DreamTech Press.
4. Mastering C++, K.R. Venugopal, Rajkumar Buyya, McGraw Hill Education, 2017.

BCAFSCO302T23	Computer Networks	*L* *3* *T* *0* *P* *0*	*Credits: 3*	*Marks: 100*

Course Outcomes:

At the completion of the course, a student will be able to:

1. Independently understand basic computer network technology.
2. Identify the different types of network topologies and protocols.
3. Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.
4. Identify the different types of network devices and their functions within a network
5. Familiarity with the basic protocols of computer networks, and how they can be used to assist in network design and implementation.

Detailed Syllabus

Unit I

Basic Concepts: Components of data communication, standards and organizations, Network Classification, Network Topologies; network protocol; layered network architecture; overview of OSI reference model; overview of TCP/IP protocol suite.

Unit II

Physical Layer: Cabling, Network Interface Card, Transmission Media Devices- Repeater, Hub, Bridge, Switch, Router, Gateway.

Data Link Layer: Framing techniques; Error Control; Flow Control Protocols; shared media protocols - CSMA/CD and CSMA/CA.

Unit III

Network Layer: Virtual Circuits and Datagram approach, IP addressing methods - Subnetting; Routing Algorithms (adaptive and non-adaptive)

Transport Layer: Transport services, Transport Layer protocol of TCP and UDP

Unit IV

Application Layer: Application layer protocols and services - Domain name system, HTTP, WWW, telnet, FTP, SMTP

Unit V

Network Security: Common Terms, Firewalls, Virtual Private Networks.

Text Books:

1. B.A. Forouzan: Data Communication and Networking, 4th Edition, Tata McGraw Hill, 2007.
2. D.E. Comer, Internetworking with TCP/IP, Vol. I, Prentice Hall of India, 1998.

Reference Books:

1. W. Stalling, Data & Computer Communication, 8th edition, Prentice Hall of India, 2006.
2. D. Bertsekas, R. Gallager, Data Networks, 2nd edition, Prentice Hall of India, 1992.

BCAFSCO303T23	Software Engineering	*L* 3	*T* 0	*P* 0	*Credits: 3*	*Marks: 100*

Course Outcomes

At the completion of the course, a student will be able to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. Communicate effectively with a range of audiences
4. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
5. Acquire and apply new knowledge as needed, using appropriate learning strategies
6. Develop an efficient software using latest tools and techniques. Use of computer aided designing and automated testing tools.

Detailed Syllabus

Unit I

Software Process: Introduction, Software Engineering Paradigm, life cycle models (water fall, incremental, spiral, evolutionary, prototyping, object oriented), System engineering, computer based system, verification, validation, life cycle process, development process, system engineering hierarchy.

Unit II

Software requirements: Functional and non-functional requirements, requirement engineering process, feasibility studies, validation and management, software prototyping, prototyping in the software process, rapid prototyping techniques, user interface prototyping, Software document analysis and modeling, data, functional and behavioral models, structured analysis and data dictionary

Unit III

Design Concepts and Principles: Design process and concepts, modular design, design heuristic, design model and document, Architectural design, software architecture, data design, architectural design, transform and transaction mapping, user interface design, user

interface design principles, Real time systems, Real time software design, system design, real time execution, data acquisition system, monitoring and control system.

Unit IV

Software Configuration Management: The SCM process, Version control, Change control, Configuration audit, SCM standards.

Software Project Management: Measures and measurements, S/W complexity and science measure, size measure, data and logic structure measure, information flow measure. Estimations for Software Projects, Empirical Estimation Models, Project Scheduling,

Unit V

Testing: Taxonomy of software testing, levels, test activities, types of software test, black box testing boundary conditions, structural testing, test coverage criteria based on data flow, mechanisms, and regression testing in the large. Software testing strategies, strategic approach and issues, Unit testing, integration testing, validation testing, system testing and debugging

Trends in Software Engineering: Reverse Engineering and Re-engineering: wrappers, Case Study of CASE tools.

Text Books:

1. Roger S. Pressman, Software engineering- A practitioner's Approach, McGraw-Hill
2. Ian Sommerville, Software engineering, Pearson education Asia, 6th edition. 2000.
3. Pankaj Jalote- An Integrated Approach to Software Engineering, Springer Verlag, 1997.

Reference Books:

1. James F Peters and Witold Pedryez, "Software Engineering-An Engineering Approach", John Wiley and Sons, New Delhi, 2000.
2. Ali Behforooz and Frederick J Hudson, "Software Engineering Fundamentals", Oxford University Press, New Delhi, 1996.

BCAFSCO304T23	**Computer Architecture**	***L T P*** ***3 0 0***	***Credits: 3***	***Marks: 100***

Course Outcomes:

At the completion of the course, a student will be able to:

1. Understand the theory and architecture of central processing unit.

2. Analyze some of the design issues in terms of speed, technology, cost, performance.

3. Design a simple CPU with applying the theory concepts.

4. Use appropriate tools to design verify and test the CPU architecture.

5. Learn the concepts of parallel processing, pipelining and interprocessor communication.

6. Understand the architecture and functionality of central processing unit.

7. Exemplify in a better way the I/O and memory organization.

8. Define different number systems, binary addition and subtraction, 2's complement representation and operations with this representation.

Detailed Syllabus

Unit-I

Basic Computer Organization and Design: Instruction Codes, Computer Registers, Common bus system, Computer Instructions, Instruction formats, Instruction Cycle, Fetch and Decode, Flowchart for Instruction cycle, Register reference instructions.

Unit-II

Register Transfer and Micro-operation: Register Transfer Language, Register Transfer, Bus and Memory Transfer, Three state bus buffers, Memory Transfer; Arithmetic Micro operations, Logic Micro operations.

Unit-Ill

Micro programmed Control Unit: Design of Control Unit, and Central Processing Unit: Introduction, General Register Organization, and Stack Organization: Register stack. Memory stack; Instruction Formats, Addressing Modes

Unit-IV

Input Output Organization: Peripheral devices, Input Output interface, Modes of Data Transfer, Priority Interrupt and Direct Memory Access.

Unit-V

Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.

Text Books:

1. M. Mano, Computer System Architecture, Pearson Education 1992.
2. Digital Design, M.M. Mano, Pearson Education Asia, 1979.

Reference Books:

3. A. J. Dos Reis, Assembly Language and Computer Architecture using C++ and JAVA, Course Technology, 2004.
4. W. Stallings, Computer Organization and Architecture Designing for Performance, 8th Edition, Prentice Hall of India, 2009

BCAFSDS301T23	Frontend Development with ReactJS	*L* *3*	*T* *0*	*P* *0*	*Credits: 3*	*Marks: 100*

Course Outcomes

At the completion of the course, a student will be able to:

1. Be familiar with client-side Javascript application development and the React library.
2. Implement single page applications in React.
3. Use various React features including components and forms.
4. Implement a functional front-end web application using React.
5. Use Reactstrap for designing responsive React applications.
6. Use Redux to design the architecture for a React-Redux application

Unit 1: Refreshing ES6

Specifications and Features, Introduction, the arrow functions, New Literal Syntax, Classes, Inheritance using extends, Default Parameter Values, Spread Operator (...), Iterators and Generators, Introduction to React, Features of React, why we Need React

Unit 2: ECMA, E6

ECMA Script, ES6 let and const, the arrow functions, New Literal Syntax, Classes, Inheritance using extends, Default Parameter Values, Spread Operator (...), Iterators and Generators, Features of React, Practical Application, Why need React, How React Works, Leveraging Virtual DOM, Setting up React

Unit 3: JSX

Why JSX, Embedding JavaScript, Expression in JSX, JSX as an Expression, Nested elements in JSX, JSX Attributes, JSX Comments, JSX Styling and representation as object, The State of the Component, Defining State, Changing the State, Props, Validation, Validators

Unit 4: Elements

Rendering Element, about render (), Creating React Element, Updating Element, components, Introducing Components, Types of Components, Functional Component, Functional Components as Stateless, Using Functional Component

Unit 5: Redux

Redux Concepts, Redux Principles, Data Flow, Actions, Functions, Reduces, Testing, DevTools, React & Redux Integrate

Text Books:

5. M. Mano, Computer System Architecture, Pearson Education 1992.
6. Digital Design, M.M. Mano, Pearson Education Asia, 1979.

Reference Books:

7. A. J. Dos Reis, Assembly Language and Computer Architecture using C++ and JAVA, Course Technology, 2004.
8. W. Stallings, Computer Organization and Architecture Designing for Performance, 8th Edition, Prentice Hall of India, 2009

BCAFSSE301T23	Database Management System	*L* 3	*T* 0	*P* 0	*Credits: 3*	*Marks: 100*

Course Outcomes:

At the completion of the course, a student will be able to:

1. Introduce the management of database systems and understand the fundamentals ofrelational systems including data models, database architectures, and database manipulations.
2. Provide knowledge about relational database model.
3. Learn the basic concepts of databases in general with an emphasis on relational databases, modelling techniques and writing queries.
4. Understand Normalization techniques, Transaction processing, Concurrency Controltechniques and Recovery of databases against crashes are also covered.

Unit I

INTRODUCTIONPurpose of Database System -– Views of data – Data Models – Database Languages –– Database System Architecture – Database users and Administrator – Entity– Relationship model (E-R model) – E-R Diagrams -- Introduction to relational databases

Unit II

RELATIONAL MODEL:The relational Model – The catalog- Types– Keys - Relational Algebra – Domain Relational Calculus – Tuple Relational Calculus - Fundamental operations – Additional Operations- SQL fundamentals, Oracle data types, Data Constraints, Column level & table Level Constraints, working with Tables, Defining different constraints on the table, Defining Integrity Constraints in the ALTER TABLE Command, Select Command, Logical Operator, Range Searching, Pattern Matching, Oracle Function, Grouping data from Tables in SQL, Manipulation Data in SQL. Joining Multiple Tables (Equi Joins), Joining a Table to itself (self Joins), Sub queries Union, intersect & Minus Clause

Unit III

SQL VIEWS, INDEX AND TRIGGERS Creating view, Renaming the Column of a view, Granting Permissions, - Updating, Selection, destroying view Creating Indexes, Creating, and managing User, Integrity – Triggers - Security – Advanced SQL features –Embedded SQL– Dynamic SQL- Missing Information– Views – Introduction to Distributed Databases and Client/Server Databases

Unit IV DATABASE DESIGN Functional Dependencies – Non-loss Decomposition – Functional Dependencies – First, Second, Third Normal Forms, Dependency Preservation – Boyce/Codd Normal Form-Multi-valued Dependencies and Fourth Normal Form – Join Dependencies and Fifth Normal Form

Unit V TRANSACTIONS:Transaction Concepts - Transaction Recovery – ACID Properties – System Recovery – Media Recovery – Two Phase Commit - Save Points – SQL Facilities for recovery –Concurrency – Need for Concurrency – Locking Protocols – Two Phase Locking – Intent Locking – Deadlock-Serializability – Recovery Isolation Levels – SQL Facilities for Concurrency.

BCAFSCO301P23	Object Oriented Programming Lab	*L* *0*	*T* *0*	*P* *4*	*Credits: 2*	*Marks: 100*

Course Outcomes:

At the completion of the course, a student will be able to:

After successful completion of the course, students will be able to:

1. Acquire profound knowledge of object oriented programming.
2. Demonstrate the difference between the solutions offered by traditional imperative problem solving method and object-oriented method
3. Explain the class inheritance, data encapsulation, polymorphism as fundamental building blocks to generate reusable code.
4. Understand and implement error handling and file handling routines.

Exercises:

1. Write a C++ program for function with default arguments
2. Write a C++ program to illustrate the concept of call by value.
3. Write a C++ program to illustrate the concept of call by reference
4. Write a C++ program to illustrate the concept of call by address.
5. Write a C++ program to illustrate the concept of Classes and objects.
6. Write a C++ program to create a mark list using arrays in C++ programming language.
7. Write a C++ program to perform operation on string class.
8. Write a C++ program to implement static member function.
9. Write a C++ program to display the details of a person using constant member function.
10. Write a C++ program to illustrate the concept of unary operator overloading.
11. Write a C++ program to illustrate the concept of Binary operator overloading.
12. Write a C++ program to illustrate the concept of function overloading.
13. Write a C++ program to multiply the positive numbers using single inheritance.
14. Write a C++ program using multiple inheritances for collecting employee details.
15. Write a C++ program for calculation of area of shapes using virtual functions.
16. Write a C++ program for a student mark list processing using virtual base class.

17. Write a C++ program using function template to find the maximum of two data.
18. Write a C++ program using class template to find the greater of the given two data's.
19. Write a C++ program for creating student data using sequential file access.
20. Write a C++ program for creating student data using random file access.

BCAFSDS301P23	Frontend Development with ReactJS Lab	*L T P* *0 0 4*	*Credits: 2*	*Marks: 100*

Course Outcomes

At the completion of the course, a student will be able to:

1. Learn the basics of JavaScript ES6
2. Understand the basics of React
3. Implementing Lists in React
4. Implementing the JavaScript and React on Real Time applications

1. Introduction to JavaScript ES6

Review key features in JavaScript ES6 that will help you code more efficiently and prepare to usethe React library.

2. Introduction to React

Start using the features of ES6 in order to build highly efficient components in React.

3. Introduction to JSX

Dive deeper into React and JSX to build basic components that enable you to create flexible user interfaces.

4. Working With Lists in React

Build your skills in React by manipulating data to display key information.

1. Working With Simple Components

Build skills with multiple components and track state by working through ATM and shopping card exercises.

2. Tic-Tac-Toe Exercise With React

Delve deeper into working with multiple components and track state by creating a tic-tac-toegame.

3. Working With React Components

Learn routing, linking, and how to style and customize components using frameworks like Bootstrap in React.

4. Bad Bank Exercise

Build a bank application using React, making decisions about routing, context, styles, parent-child components, tracking state, forms, and handling events.

5. Shopping Cart Exercise

Develop a complete application with a store, shopping cart, and checkout.

6. Testing

Use the full set of skills learned in the course to build and deploy an application on Amazon Web Services, and then learn how to test your code using the React library.

7. GitHub Portfolio Project

The culmination of all your learnings, you'll work on the front-end of a banking app that you can share with potential employers on your GitHub portfolio

BCAFSSE301P23	Database Management Lab	*L* *0*	*T* *0*	*P* *4*	*Credits: 2*	*Marks: 100*

Course Outcomes

At the completion of the course, a student will be able to:

1. Provide practical experience to students and reinforce the theory concepts
2. To learn and understand the fundamental of Oracle Database.
3. To implement the various queries practically.

LIST OF PROGRAMS:

1. Create User in Oracle Database and grant and revoke the privileges and use of commit save point rollback command.
2. Create the following:
   Synonym sequences and Index
   Create alter and update views.
3. Create PL/SQL program using cursors, control structure, exception handling
4. Create following:
   Simple Triggers
   Package using procedures and functions.
5. Create the table for
   COMPANY database

STUDENT database and insert five records for each attribute.

6. Illustrate the use of SELECT statement
7. Conditional retrieval - WHERE clause
8. Query sorted - ORDER BY clause
9. Perform following:
   UNION, INTERSECTION and MINUS operations on tables.
   UPDATE, ALTER, DELETE, DROP operations on tables
10. Query multiple tables using JOIN operation.
11. Grouping the result of query - GROUP BY clause and HAVING clause
12. Query multiple tables using NATURAL and OUTER JOIN operation.
13. Create a scenario based on real time domain

SEM IV

BCAFSCO401T23	Python Programming	L T P 3 0 0	Credits: 3	Marks: 100

Course Outcomes:

At the completion of the course, a student will be able to:

1. Students will be able to understand why Python is a useful scripting language for developers.
2. To acquire knowledge and programming skills in python to solve problems in different domains.
3. Students will learn how to design and program Python applications.

UNIT I

Basics of Python:Python Installation and Working of it, get familiar with python variables and data types, Operator understanding and its usage, detail study of python blocks,

UNIT II

Structure Types and mutability :Hands on with conditional blocks using if, else and elif, Hands on examples and study of looping with range, list, and dictionaries. hands on to organize python code with function, modular approach in python.

UNIT III

Exception, Testing and Debugging Handling if exceptions to handle the code cracks, handling and helping file operations, coding with the exceptional handling, and testing Anonymous method, Properties, Indexers, Exception Handling

UNIT IV

Classes and OOP Concepts :Procedural and Object-Oriented Programming, Classes and working with instances, Method overloading, Polymorphism, importing internal module as well as external modules in the code Packages understanding and their usage, hands on with Lamba function in python coding with the use of functions, modules, and external packages

UNIT V

Algorithm and Data Structure Stack, Queue, Tree, ordered list, Introduction to Recursion, Divide and Conquer Strategy, Greedy Strategy, Graph Algorithms. Advance Topics: Regular Expression, Multi thread Programming, Security.

Textbooks and Reference Books:

1. Starting Out with Python (2009) Pearson, Tonny Gaddis
2. Beginning Pyhton Wrox Publication Peter Norton, Alex Samuel
3. Python Algorithms Apress, Magnus Liet Hetland,

Essential Reading / Recommended Reading:

1. Python Object Oriented Programming PACKT Press, Dusty Phillips
2. Python for Unix and Linux System Administration O'Relly, Noad Gift

BCAFSCO402T23	Backend Development with NodeJS	L T P 3 0 0	Credits: 3	Marks: 100

Course Outcomes:

At the completion of the course, a student will be able to:

1. Introduction to Node.js and how is it beneficial in website and applications
2. How to build modules using Node.js
3. To understand and learn the buffers
4. To learn about REST API

Unit 1

Introduction to Node.js :What is Node.js, History of Node.js, Why Node.js, Node.js Architecture, Working and Features, Installation and Setup, Installing Node.js, Launching REPL, Environment, Installing Visual Studio, Code Editor, Components of Node.js

Unit 2

Modules :Module Exports, Export Object, Export Functions, Export Functions as Class, loading module from, Separate Folder, Modules, File System Module, Reading and Writing into, Files, Appending and Opening Files, Events and Event Emitters, Handling Events, Customized Class for Handling Events, In-built Modules File Systems, Operating System

Unit 3

Buffers :Writing to Buffers, reading from Buffers, Concatenating Buffers, Copying Buffers, Slicing Buffers, The Stream Module, reading from Stream, Writing to Stream, Pipes, Pipe Chaining

Unit 4

REST API :Explain REST API, Describe Node.js express, Discuss the importance of express, Explain the installing process of express, learn express request and response, Describe routing, REST API: Intro to API, History of API Development, Development of AJAX, CRUD

Unit 5: Project

Text Books:

Text Books:

1. "Node.js in Action" by Mike Cantelon, Marc Harter, and T.J. Holowaychuk
2. "Learning Node.js: A Hands-On Guide to Building Web Applications in JavaScript" by Marc Wandschneider
3. "Node.js, MongoDB, and AngularJS Web Development" by Brad Dayley
4. "Node.js Design Patterns" by Mario Casciaro and Luciano Mammino

Reference Books:

1. "Node.js Web Development, Fourth Edition" by David Herron
2. "Learning Node.js Development" by Andrew Mead
3. "Professional Node.js: Building JavaScript Based Scalable Software" by Pedro Teixeira

BCAFSCO403T23	Programming in Java	*L* *3*	*T* *0*	*P* *0*	*Credits: 3*	*Marks: 100*

Course outcomes:

At the completion of the course, a student will be able to:

1. Recall the basic knowledge on Object Oriented concepts specially in java.
2. Create & design applications using Object Oriented Programming Concepts using java
3. Describe for compile, test and run Java programs comprising more than one class
4. Create simple data structures like arrays in a Java program
5. Explain members of classes found in the Java API

Detailed Syllabus

Unit I

Introduction to Java: Features of Java, JDK Environment, Object Oriented Programming Concept Overview of Programming, Paradigm, Classes, Abstraction, Encapsulation, Inheritance, Polymorphism, Difference between C++ and JAVA

Unit II

Java Programming Fundamental: Structure of Java program, Data types, Variables, Operators, Keywords, Naming Convention, Decision Making (if, switch),Looping(for, while) ,Type Casting Classes and Objects: Creating Classes and objects, Memory allocation for objects, Constructor

Unit III

Arrays and Strings: Arrays, Creating an array, Types of Arrays, String class Methods, String Buffer

methods.

Implementation of Inheritance, Implementation of Polymorphism, Method Overloading, Method Overriding, Nested and Inner classes

Unit IV

Abstract Class, Interface and Packages: Modifiers and Access Control, Abstract classes and methods, Interfaces, Packages Concept, Creating user defined packages

Exception Handling: Exception types, using try catch and multiple catch, Nested try, throw, throws and finally, Creating User defined Exceptions.

Unit V

File Handling: Byte Stream, Character Stream, File 10 Basics, File Operations, Creating file, Reading file, Writing File

Applet Programming: Introduction. Types Applet, Applet Life cycle, Creating Applet, Applet tag

Text Books:

1. Herbert Schildt, Java 7, The Complete Reference, 8th Edition, 2009.
2. E Balagurusamy, Programming with JAVA, TMH, 2007.

ReferneceBooks:

1. Ivan Bayross, Web Enabled Commercial Application Development Using Html, Dhtmljavascript, Perl Cgi, BPB Publications, 2009.
2. Cay Horstmann, BIG Java, Wiley Publication , 3rd Edition., 2009.

BCAFSDS401T23	Cloud Computing	L T P 3 0 0	Credits: 3	Marks: 100

Course outcomes:

At the completion of the course, a student will be able to:

1. Remember concept of Cloud Computing, benefit and challenges associated with it.
2. Understand various cloud services, cloud service providers and frameworks being used.
3. Describe importance of virtualization along with their technologies.
4. Identify the need for the virtualization and advantage and limitations of using virtualization concept.
5. Analyze the open stack & Google Cloud platform components and understand Mobile Cloud Computing
6. Investigate the security aspect in cloud, standards for security framework, challenges.

Detailed Syllabus:

UNIT I

Cloud Introduction; Cloud Computing Fundamentals: Cloud Computing definition, Types of Cloud, Cloud services: Benefits and challenges of cloud computing, Evolution of Cloud Computing usage scenarios and Applications, Business models around Cloud Major Players in Cloud Computing.

UNIT II

Cloud Services and File System: Types of Cloud services, Service providers, Google App Engine, Amazon EC2, Microsoft Azure, Sales force. Introduction to Map Reduce, GFS, HDFS, Hadoop Framework,

UNIT III

Collaborating With Cloud: Collaborating on Calendars, Schedules and Task Management **Collaborating** on Event Management, Contact Management, Collaborating via Web-Based Communication Tools, Evaluating Web Mail Services and Collaborating via Social Networks

UNIT IV

Virtualization for Cloud: Need for Virtualization - Pros and cons of Virtualization, Types of Virtualization, System VM, Process VM, Virtual Machine monitor.

UNIT V

Security, Standards, and Applications

Security in Clouds: Cloud security challenges, Common Standards: The Open Cloud Consortium - The Distributed management Task Force - Standards for application Developers Standards for Messaging - Standards for Security. End user access to cloud computing, Mobile Internet devices and the cloud.

Text Books:

1. Bloor R., Kanfman M., Halper F. **Judith** Hurwitz "Cloud Computing " Wiley India Edition, 2010
2. John Rittinghouse & James Ransome, "Cloud Computing Implementation Management and Strategy", CRC Press, 2010
3. Antony T Velte, Cloud Computing : "A Practical Approach", McGraw Hill,2009

Reference Books:

1. Michael Miller, Cloud Computing: "Web-Based Applications That Change the Way You Work and Collaborate Online", Que Publishing, August 2008.
2. James E Smith, Ravi Nair, "Virtual Machines", Morgan Kauftnann Publishers, 2006.

Online Reading/Supporting Material

1. Haley Beard, "Cloud Computing Best Practices for Managing and Measuring Processes for On-demand Computing", Applications and Data Centers in the Cloud with SLAs, Emereo Pty Limited, July 2008.
2. webpages.iust.ac.ir/hsalimi/.../89.../Cloud%20Common%20standards.pptop ennebula.org,
3. www.cloudbus.org/cloudsim/, http://wvvw.eucalyptus.com/
4. hadoop.apache.org
5. http://hadoop.apaehe.org/docs/stable/hdfs_design,html

BCAFSSE401T23	E-Commerce Technologies	*L* *3*	*T* *0*	*P* *0*	*Credits: 3*	*Marks: 100*

Course outcomes:

At the completion of the course, a student will be able to:

1. Identify and explain fundamental web site tools including design tools, programming tools, and data processing tools.

2. Apply the solutions on finding major electronic payment issues and options.
3. Acquire the knowledge of security issues and explain procedures used to protect against security threats.
4. Communicate effectively in ways appropriate to the discipline, audience and purpose.
5. Implement the corrective measures to management issues underlying e-Commerce issues including organizational structure, strategic planning, goal setting, corporate social responsibility, international arena, changing market intermediaries, resource allocation and customer service.

Detailed Syllabus

Unit I

An introduction to Electronic commerce: What is E-Commerce (Introduction and Definition), Main activities E-Commerce. Goals of E-Commerce, Technical Components of E-Commerce, Functions of E-Commerce, Advantages and disadvantages of E-Commerce, Scope of E-Commerce, Electronic Commerce Applications, 9 Electronic Commerce and Electronic Business(C2C)(C2G;G2G, B2G, B2P, B2A, P2P, B2A, C2A, B2B, B2C)

Unit II

The Internet and WWW: Evolution of Internet, Domain Names and Internet Organization (.edu, .com, .mil, .gov, .net etc.) , Types of Network, Internet Service Provider, World Wide Web, Internet & Extranet, Role of Internet in B2B Application, building own website, Cost, Time, Reach, Registering a Domain Name, Web promotion, Target email, Barter, Exchange, Shopping Bots

Unit III

Internet Security: Secure Transaction, Computer Monitoring, Privacy on Internet, Corporate Email privacy, Computer Crime(Laws , Types of Crimes), Threats, Attack on Computer System, Software Packages for privacy, Hacking, Computer Virus(How it spreads, Virus problem, virus protection, Encryption and Decryption, Secret key Cryptography, DES, Public Key Encryption, RSA, Authorization and Authentication, Firewall, Digital Signature(How it Works)

Unit IV

Electronic Data Exchange: Introduction, Concepts of EDI and Limitation, Applications of EDI, Disadvantages of EDI, EDI model, Electronic Payment System: Introduction, Types of Electronic Payment System, Payment Types, Value Exchange System, Credit Card System, Electronic Fund Transfer, Paperless bill, Modern Payment Cash, Electronic Cash

Unit V

Planning for Electronic Commerce: Planning Electronic Commerce initiates, Linking objectives to business strategies, Measuring cost objectives, Comparing benefits to Costs, Strategies for developing electronic commerce web sites.

Internet Marketing; The PROS and CONS of online shopping, The cons of online shopping. Justify an Internet business, Internet marketing techniques, The E-cycle of Internet marketing, Personalization e-commerce.

Text Books:

1. G.S.V.Murthy, E-Commerce Concepts, Models, Strategies- :- Himalaya Publishing House, 2011.
2. Kamlesh K Bajaj and Debjani Nag, E- Commerce, 2005.

Reference Books:

3. Gray P. Schneider, Electronic commerce, International Student Edition, 2011.
4. E-Commerce, Fundamentals and Applications, Wiely Student Edition.

BCAFSPPCO401P23	**Python programming Lab**	***L T P*** ***0 0 4***	***Credits: 2***	***Marks: 100***

Course Outcomes:

At the completion of the course, a student will be able to:

1. Understand the basic syntax of Python
2. Learn different searching and sorting methods
3. Implement the External & Internal Library
4. Implement OOP concept using Python

List of Programs

1. Implement a Python program to Calculate GCD of two numbers.
2. Implement a Python Program to Calculate the square root of a number by Newton's Method.
3. Implement a Python program to Calculate the exponentiation of a number.
4. Implement a Python Program to Calculate the maximum from a list of numbers.
5. Implement a Python Program to perform Search

6. Implement a Python Program to perform Liner search
7. Implement a Python Program to perform Binary search
8. Implement a Python Program to perform insertion sort.
9. Implement a Python Program to perform selection sort.
10. Implement a Python program to multiply matrices.
11. Implement a Python program to Calculate the most frequent words in a text read from a file.
12. Implement function overloading with different function signatures.
13. Implement concept of class, instances, and inheritance.
14. Implement internal and external library.
15. Solve algorithmic problems by program using different problem-solving strategies.
16. Search content using regular expression library in python.
17. Implement Matrix multiplication using multi-threading in python.
18. Perform Linux administration task using python.

BCAFSCO402P23	**Backend Development with NodeJS Lab**	***L T P*** ***0 0 4***	***Credits: 2***	***Marks: 100***

At the completion of the course, a student will be able to:

Course Outcomes:

1. Understand the concept of REPL shell
2. Learn the Global objects in Node.js
3. Learn to develop HTTP servers and clients
4. Learn about Streams , events & pipes

List of Exercises:

1.Non-blocking IO. Event loop. Using callbacks and promises. Continuation-passing style in JavaScript (CPS) and Node.js. (1 astr. h.)

2.Node.js and npm – installation, packages, command line arguments. Using Visual Studio Code JS / ES6 editor, linting. (1 astr. h.)

3.Using Node.js shell (REPL). Running scripts. REPL shell – main features, modes and commands, defining custom commands. (1 astr. h.)

4.Modules and module usage patterns, require, core modules. (2 astr. h.)

5.Global objects in Node.js – Buffer, dirname, filename, console, exports, global, module, process, setImmediate(), setInterval(), setTimeout(). (1 astr. h.)

6.Developing HTTP servers and clients using HTTP module – developing hello-world web server. Routing requests. HTTP methods. Developing HTTP clients using http.get() and http.request(). Creating custom modules. (4 astr. h.)

7.Serving files (File System module) – asynchronous vs. synchronous, completion callbacks, using promises. Buffers. Main file and directory operations, getting read/write streams, statistics, watching file-system for changes. (2 astr. h.)

8.Streams, events and pipes – EventEmitter, custom events. Readable, Writable, Duplex and Transform streams. Examples – HTTP requests / responses, fs write streams, zlib streams, stdin, stdout. Connecting streams using pipe(). Main type of stream events – data, end, error, finish, drain, readable, pipe, unpipe. Implementing custom readable, writable and transform streams by extending base classes (ES5, ES6) and using Simplified Constructor approach. (4 astr. h.)

9.Process and child_process – arguments, events, exit codes, promise rejection and exception handling, benchmarking, spawn(), exec(), fork(), event-driven inter-process communication (IPC). (2 astr. h.)

10.Final test + practical problem + discussion of additional questions. (2 astr. h.)

BCAFSCO403P23	**Java Programming Lab**	***L***	***T***	***P***	***Credits: 2***	***Marks: 100***
		0	***0***	***4***		

Course outcomes:

At the completion of the course, a student will be able to:

1. Recall traditional imperative design and object-oriented Design using java
2. Understand class structures as fundamental, modular building blocks
3. Describe for compile, test and run Java programs comprising more than one class
4. Create simple data structures like arrays in a Java program
5. Create and Specify classes found in the Java API

Exercises:

1. Write a program to find the largest of n natural numbers.
2. Write a program to find whether a given number is prime or not.
3. Write a menu driven program for following:
   a. Display a Fibonacci series
   b. Compute Factorial of a number
   c. Write a program to check whether a given number is odd or even.
   d. Write a program to check whether a given string is palindrome or not.
4. Write a program to print the sum and product of digits of an Integer and reverse the Integer.
5. Write a program to create an array of 10 integers. Accept values from the user in that array. Input another number from the user and find out how many numbers are equal to the number passed, how many are greater and how many are less than the number passed.
6. Write a program that will prompt the user for a list of 5 prices. Compute the average of the prices and find out all the prices that are higher than the calculated average.
7. Write a program in Java to input N numbers in an array and print out the Armstrong numbers from the set.
8. Write Java program for the following matrix operations:
   a. Addition of two matrices
   b. Summation of two matrices
   c. Transpose of a matrix
   d. Input the elements of matrices from user.
9. Write a Java program that computes the area of a circle, rectangle and a Cylinder using function overloading.
10. Write a Java for the implementation of multiple inheritance using interfaces to calculate the area of a rectangle and triangle.

11. Write a Java program to create a frame window in an Applet. Display your name, address and qualification in the frame window.

12. Write a Java program to draw a line between two coordinates in a window.

13. Write a java program to display the following graphics in an applet window.

    a. Rectangles

    b. Circles

    c. Ellipses

    d. Arcs

    e. Polygons

14. Write a program that reads two integer numbers for the variables a and b. If any other Character except number (0-9) is entered then the error is caught by Number Format Exeeption object. After that ex.getMessage() prints the information about the error occurring causes.

15. Write a program for the following string operations:

    a. Compare two strings

    b. Concatenate two strings

    c. Compute length of a string

16. Create a class called Fraction that can be used to represent the ratio of two integers. Include appropriate constructors and methods. If the denominator becomes zero, throw and handle an exception.

BCAFSDS401P23	**Cloud Computing Lab**	**Course Outcomes**	**Course Outcomes**	**Course Outcomes**

Course Outcomes:

At the completion of the course, a student will be able to:

1. Define & implement Virtualization using different types of Hypervisors

2. Experience storage, calendar and document editing services offered by Google cloud.
3. Describe the functioning of Platform as a Service
4. Explore the Microsoft cloud platform.
5. Analyze and understand the functioning of different components involved in Amazon web services cloud platform.
6. Design & Synthesize Storage as a service using own Cloud

Exercises:

1 Create virtual machines that access different programs on same platform,

2 Create virtual machines that access different programs on different platforms.

3 Exploring Google cloud for the following

a) Storage

b) Sharing of data

c) manage your calendar, to-do lists,

d) a document editing tool

4 Exploring Microsoft cloud
Exploring Amazon cloud

SEM V

BCAFSCO501T23	Andriod Programming	*L* *3*	*T* *0*	*P* *0*	*Credits: 3*	*Marks: 100*

Course outcomes:

At the completion of the course, a student will be able to:

1. Remember Android platform, Architecture and features.
2. Understand the User Interface and develop activity for Android App.
3. Design and implement Database Application and Content providers.
4. Use multimedia, camera and Location based services in Android App.
5. Discuss various security issues in Android platform

Detailed Syllabus

UNIT I

Introduction: History of Android, Introduction to Android Operating Systems, Android Development Tools, Android Architecture.

UNIT II

Overview of object oriented programming using Java: OOPs Concepts: Inheritance, Polymorphism,

Interfaces, Abstract class, Threads, Overloading and Overriding, Java Virtual Machine

UNIT III

Development Tools: Installing and using Eclipse with ADT plug-in, Installing Virtual machine for Android sandwich/Jelly bean (Emulator), configuring the installed tools, creating a android project, Hello World, run on emulator, Deploy it on USB-connected Android device.

UNIT IV

User Interface Architecture: Application context, intents. Activity life cycle, multiple screen sizes.

User Interface Design: Form widgets, Text Fields, Layouts. Button control, toggle buttons, Spinners (Combo boxes), Images, Menu, and Dialog.

UNIT V

Database: Understanding of SQLite database, connecting with, the database.

Text Books:

1. Android application development for java programmers. By James C. Sheusi. Publisher: Cengage Learning, 2013.
2. Android Wireless Application Development By Lauren Darcey and Shane Conder, Pearson Education, 2nd ed. (2011)
3. Using SQLite By Jay A. Kreibich, Publisher: O'Reilly Media.

Reference Books:

1. Mobile Computing using Android and iPhone [ISBN: 978-93-81786-93-2] by Bharat & Company.
2. Professional Android 2 Application Development Reto Meier, Wiley India Pvt Ltd (2011).
3. Beginning Android Mark L Murphy, Wiley India Pvt Ltd.

Online Reading / Supporting Material:

1. http://www.developer.android.com
2. http://developer.android.com/aboLJt/versions/indexJitml
3. http://developer.android.com/training/basics/flrstapp/indexJ^tml
4. http://docs.oracle.com/javase/tutorial/index.htm (Available in the form of free downloadable ebooks also).

BCAFSCO502T23	Software Testing Concepts	L 3	T 0	P 0	Credits: 3	Marks: 100

Course Outcomes:

At the completion of the course, a student will be able to:

1. Learn systematic approach to the development, operation, maintenance, and retirement of software.
2. Learn how to use available resources to develop software, reduce cost of software and how to maintain quality of software.

3. Illustrate Methods and tools of testing and maintainace of software's.
4. Investigate the reason for bugs and analyze the principles in software testing to prevent and remove bugs.
5. Develop methods and procedures for software development that can scale up for large systems and that can be used to consistently produce high-quality software at low cost and with a small cycle time.

Detailed Syllabus

UNIT I

Introduction

Strategic Approach to Software Testing, Test Strategies for Conventional Software, Validation Testing, System Testing, Basic Terminologies, V Shaped Software Lifecycle Model.

UNIT II

Functional Testing\ Black-box Testing

Boundary Value Analysis, Equivalence Class Testing, Decision Table Based Testing

UNIT III

Structural Testing: White-box Testing

Basis Path Testing: Program Graph, DD Path graph, Cyclomatic Complexity, Graph Matrices, Control Flow Testing: Statement Coverage, Branch Coverage, Condition Coverage, Path Coverage.

UNIT IV

Software Quality Assurance, test optimization, Eleven Step Testing Process and Testing Security.

UNIT V

Software Reusability, Software Metrics, Software Testing Tools, Defect Tracking Tools, Defect Management Tools and Challenges

Books Recommended;

1. Roger S. Pressman, Software Engineering: A Practitioner's Approach, Seventh Edition, Mc Graw Hill Education.2009.
2. Yogesh Singh, Software Testing, Cambridge University Press, 2011.

BCAFSCO503T23	Data Science Project Management	*L T P* *3 0 0*	*Credits: 3*	*Marks: 100*

Course Outcomes:

1. Identify the need for data science and solve basic problems using Python built-in data types and their methods.
2. Employ efficient storage and data operations using NumPy arrays.
3. Apply powerful data manipulations using Pandas.
4. To outline the need for Project Management
5. To highlight different techniques of activity planning

Unit-1

Introduction to Data Science - Why Python? - Essential Python libraries - Python Introduction- Features, Identifiers, Reserved words, Indentation, Comments, Built-in Data types and their Methods: Strings, List, Tuples, Dictionary, Set - Type Conversion- Operators. Decision Making- Looping- Loop Control statement- Math and Random number functions. User defined functions - function arguments & its types.

Practical Component:

1. Implement basic Python programs for reading input from console.
2. Perform Creation, indexing, slicing, concatenation and repetition operations on Python built-in data types: Strings, List, Tuples, Dictionary, Set
3. Solve problems using decision and looping statements.
4. Apply Python built-in data types: Strings, List, Tuples, Dictionary, Set and their methods to solve any given problem
5. Handle numerical operations using math and random number functions
6. Create user-defined functions with different types of function arguments.

Unit-2

NumPy Basics: Arrays and Vectorized Computation- The NumPy ndarray- Creating ndarrays- Data Types for ndarrays- Arithmetic with NumPy Arrays- Basic Indexing and Slicing - Boolean Indexing-Transposing Arrays and Swapping Axes. Universal Functions: Fast Element-Wise Array Functions- Mathematical and Statistical Methods-SortingUnique and Other Set Logic.

Practical Component: 1. Create NumPy arrays from Python Data Structures, Intrinsic NumPy objects and Random Functions.

2. Manipulation of NumPy arrays- Indexing, Slicing, Reshaping, Joining and Splitting.

3. Computation on NumPy arrays using Universal Functions and Mathematical methods.

4. Import a CSV file and perform various Statistical and Comparison operations on rows/columns.

5. Load an image file and do crop and flip operation using NumPy Indexing

Unit-3

Introduction to pandas Data Structures: Series, DataFrame, Essential Functionality: Dropping Entries Indexing, Selection, and Filtering- Function Application and Mapping- Sorting and Ranking. Summarizing and Computing Descriptive Statistics- Unique Values, Value Counts, and Membership. Reading and Writing Data in Text Format.

Practical Component:

1. Create Pandas Series and DataFrame from various inputs.

2. Import any CSV file to Pandas DataFrame and perform the following:

(a) Visualize the first and last 10 records

(b) Get the shape, index and column details

(c) Select/Delete the records(rows)/columns based on conditions.

(d) Perform ranking and sorting operations.

(e) Do required statistical operations on the given columns.

(f) Find the count and uniqueness of the given categorical values.

(g) Rename single/multiple columns

Unit-4

Objectives of Project Management- Importance of Project Management- Types of Projects Project Management Life Cycle- Project Selection(Data Science) – Feasibility study: Types of feasibility Steps in feasibility study

Unit-5

Setting a base line- Project management Information System – Indices to monitor progress. Importance of Contracts in projects- Teamwork in Project Management - Attributes of a good project team – Formation of effective teams – stages of team formation.

Text Books:

1. Y. Daniel Liang, “Introduction to Programming using Python”, Pearson,2012.

2. Wes McKinney, “Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython”, O’Reilly, 2nd Edition,2018.

3. Jake VanderPlas, “Python Data Science Handbook: Essential Tools for Working with Data”, O’Reilly, 2017.

Reference Books:
1. Wesley J. Chun, “Core Python Programming”, Prentice Hall,2006.
2. Mark Lutz, “Learning Python”, O’Reilly, 4th Edition, 2009.

BCAFSSE501T23	Artificial Intelligence & Machine learning	*L* 3 *T* 0 *P* 0	*Credits: 3*	*Marks: 100*

Course Outcomes:

At the completion of the course, a student will be able to:

- Demonstrate fundamental understanding of the history of artificial intelligence (AI) and its foundations.
- Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.
- Demonstrate awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.

Demonstrate proficiency developing applications in an 'AI language', expert system shell, or data mining tool**.**

Unit 1:

What is intelligence? Foundations of artificial intelligence (AI). History of AI; Problem Solving-Formulating problems, problem types, states and operators, state space, search strategies.

Unit 2:

INFORMED SEARCH STRATEGIES - Best first search, A* algorithm, heuristic functions, Iterative deepening A*(IDA), small memory A*(SMA), Game playing - Perfect decision game, imperfect decision game, evaluation function, alpha-beta pruning

Unit 3: Representation, Inference, Propositional Logic, predicate logic (first order logic), logical reasoning, forward chaining, backward chaining; AI languages and tools **-** Lisp, Prolog, CLIPS.

Unit 4: Planning- Basic representation of plans, partial order planning, planning in the blocks world, hierarchical planning, conditional planning, representation of resource constraints, measures, temporal constraints

Unit 5: Basic probability, Bayes rule, Belief networks, Default reasoning, Fuzzy sets and fuzzy logic, Decision making- Utility theory, utility functions, Decision- theoretic expert systems. Decision trees, rule-based learning, current-best-hypothesis search, least-commitment search

Unit 6: Neural networks, reinforcement learning, genetic algorithms. Communication - Communication among agents, Natural language processing, formal grammar, parsing, grammar

Reference Books:

1. Stuart Russell and Peter Norvig. Artificial Intelligence – A Modern Approach, Pearson Education Press, 2001.
2. Kevin Knight, Elaine Rich, B. Nair, Artificial Intelligence, McGraw Hill, 2008.
3. George F. Luger, Artificial Intelligence, Pearson Education, 2001.
4. Mils J. Nilsson, Artificial Intelligence: A New Synthesis, Morgan Kauffman, 2002.

BCAFSDS501T23	**Programming in .NET**	***L T P*** ***3 0 0***	***Credits: 3***	***Marks: 100***

Course Outcomes

At the completion of the course, a student will be able to:

1. Remember the programming skills and be familiar with programming environment.
2. Apply the concept so the students will be able to use ASP.NET controls in web applications.
3. Interpret the to debug and deploy ASP.NET web applications
4. Describe to create database driven ASP.NET web applications and web services

5. To develop, implement, and demonstrate Component Services, Threading, remoting, Windows services.
6. Create and develop Assemblies and Deployment in .NET, Application Development.

Detailed Syllabus

Unit I

The Framework of .Net: Building blocks of .Net Platform (the CLR, CTS and CLS), Features of .Net, deploying the .Net Runtime, Architecture of .Net platform, Introduction to namespaces & type distinction. Types & Object in .Net, the evolution of Web development.

Unit II

Class Libraries in .Net, Introduction to Assemblies & Manifest in .Net, Metadata & attributes. Introduction to C#: Characteristics of C#, Data types: Value types, reference types, default value, constants, variables, scope of variables, boxing and unboxing.

Unit III

Operators and expressions: Arithmetic, relational, logical, bitwise, special operators, evolution of expressions, operator precedence & associativity. Control constructs in C#: Decision making, loops. Classes & methods: Class, methods, constructors, destructors, overloading of operators & functions.

UNIT-IV

Inheritance & polymorphism: visibility control, overriding, abstract class & methods, sealed classes & methods, interfaces.

UNIT-V

Advanced features of C#: Exception handling & error handling, automatic memory management, Input and output (Directories, Files, and streams).

Text Books:

1. Introduction to C# using .NET By Robert J. Oberg, PHI, 2002,
2. Programming in C# by E. Balaguruswamy, Tata McGraw Hill,

Reference Books:

1. The Complete Guide to C# Programming by V. P. Jain.

BCAFSCO501P23	Android Programming Lab & Minor Project	*L* *0*	*T* *0*	*P* *4*	*Credits: 2*	*Marks: 100*

Course outcomes:

At the completion of the course, a student will be able to:

1. Students learn to develop android based applications.
2. Understand the User Interface and develop UI for Android App.
3. Design and program Database Application and Content providers using android.
4. Students learn to develop small projects using the various technologies.

Android Experiment List :

1. Create "Hello World" application that will display "Hello World" in the middle or the screen in the emulator. Also display "Hello World" in the middle of the screen in the Android Phone.
2. Create an application with login module. (Check username and password).
3. Create spinner with strings taken from resource folder (res » value folder) and on changing the spinner value, Image will change.
4. Create a menu with 5 options and selected option should appear in text box.
5. Create a list of all courses in your college and on selecting a particular course teacher-in charge of that course should appear at the bottom of the screen.
6. Create an application with three option buttons, on selecting a button colour of the screen will change.
7. Create and Login application as above. On successful login, pop up the message,
8. Create an application to Create, Insert, Update, Delete and retrieve operation on the database.

List of some minor projects during the V Semester:

- Assignment Management System
- Project ATM Banking System
- Project Cab Management System

- Project Cargo Management System
- Project City Bus Management System
- Project Civil Registration System
- Project Content Management System
- Project Customer Relationship Management System
- Project Drug Management System
- Project Employee Leave Management System
- Project Factory Information Management System
- Project Hospital Management System
- Project Infrastructure Management System
- Project Life Insurance Management System
- Project Mobile Shop
- Project Newspaper Ad Management System
- Project Online IT Service Help Desk
- Project Online Job Portal System
- Project Online Student Management System
- Project Online Tax Information System
- Project Online Voting System
- Project Patient Management System
- Project Payroll Management System
- Project School Management System
- Project Task Management System
- Project Vehicle Insurance Management System

BCAFSSE501P23	**Artificial Intelligence & Machine learning Lab**	***L T P*** ***0 0 4***	***Credits: 2***	***Marks: 100***

Course Outcomes

At the completion of the course, a student will be able to:

1. To understand the importance of supervised learning in multivariate data sets.
2. To understand the estimation procedure for multiple regression coefficients
3. To understand the assumptions in estimating regression coefficients using OLS method.
4. To implement multiple linear regression
5. To Performance analysis of Classification Algorithms on a specific dataset
6. To Implement Logistic Regression

1. Write a python program to compute Central Tendency Measures: Mean, Median, Mode, Measure of Dispersion: Variance, Standard Deviation
2. Study of Python Basic Libraries such as Statistics, Math, Numpy and Scipy.
3. Study of Python Libraries for ML application such as Pandas and Matplotlib
4. Write a Python program to implement Simple Linear Regression
5. Implementation of Multiple Linear Regression for House Price Prediction using sklearn
6. Implementation of Decision tree using sklearn and its parameter tuning
7. Implementation of KNN using sklearn
8. Implementation of Logistic Regression using sklearn
9. Implementation of K-Means Clustering
10. Performance analysis of Classification Algorithms on a specific dataset (Mini Project)

BCAFSDS501P23	**.NET Programming Lab**	***L T P*** ***0 0 4***	***Credits: 2***	***Marks: 100***

Course Outcomes

At the completion of the course, a student will be able to:

1. Apply the concept so the students will be able to use ASP.NET controls in web applications.
2. Interpret the debug and deploy ASP.NET web applications.
3. Develop, implement, and demonstrate Component Services, Threading, remoting, Windows services.
4. Identify Security in the .NET framework and Deployment in the .NET.
5. Develop Assemblies and Deployment in .NET, Application Development

Exercises:

1. Write a simple program in C# to write a string on the screen
2. Write a program in C# to prompt the user for some input and then take some action.
3. Write a program in C# to demonstrate different kinds of arrays including jagged arrays.
4. Write a program to demonstrate boxing.
5. Write a program to demonstrate how unary operators are used.
6. Write a program in C# to demonstrate how binary operators work.
7. Write a program in C# to find out trie range of number from 1-10 or 11-20 or 21-30 or less than 1.
8. Write a program in C# to find out the number entered between 1 -3.
9. Write a program in C# to override a method which calculates pay of an employees to take bonus into account.
10. Write a program in C# to ask a user to enter a choice to add, delete, modify or view address using methods for each functionality.
11. Write a program in C# to demonstrate and verify that the static constructor runs only one time, even though two instances of Class are created, and that it runs before the instance constructor runs.

BCAFSCO601P23	**Major Project/Industrial Training**	***L T P*** ***0 0 20***	***Credits: 10***	***Marks: 100***

Course outcomes:

At the completion of the course, a student will be able to:

1. Identify the real world problems and challenges that need IT based solutions and create very precise specifications of the IT solution to be designed.
2. Understand project characteristics and various stages of a project.
3. Identify project goals, constraints, deliverables, performance criteria, control needs, and resource requirements in consultation with stakeholders.
4. Implement project management knowledge, processes, lifecycle and the embodied concepts, tools and techniques in order to achieve project success.
5. Demonstrate an ability to work in teams and manage the conduct of the research study.

As such, during the development of the project students shall involve themselves in all the stages of the System Development Life Cycle (SDLC) like requirements analysis, systems design, software development/coding;, testing and documentation, with an overall emphasis on the development of reliable software systems. Since, the project work spans over the entire final semester, the students shall be advised to take up projects for solving problems of software industry or any research organization or the real life problems suggested by the faculty in-charge of BCA project work in the institutions.

This option is to be offered only in 6th Semester. The students will be allowed to work on any project based on the concepts studied in core/elective or skill based elective courses. The group size should be maximum of three (03) students. Each group will be assigned a teacher as a supervisor who will handle both their theory as well lab classes. Theory classes will cover project management techniques.

During the project work, its progress will be monitored, on monthly basis, by the internal guide. 1 Copies of Project Report to be submitted to SC (I copy to be retained by SC and 1 copy to be forwarded to Department). End Examination shall be based on Project Report, Presentation, Viva, and Demonstration of the software.

1 Duration (for 1 group),

* Presentation 20 minutes
* Viva 15 minutes
* Demonstration 15 minutes
* Report checking 10 minutes Students shall be given 30 minutes time to make preparations of their presentation and demonstration in the Lab (students are advised to carry out preparation on the previous day).

2 Project Work carries Total 300 Marks

* Project Report 100 marks
* Presentation (PowerPoint based) 80 marks
* Viva 60 marks
* Demonstration 60 marks 8.

3 Format of Project Report The manuscript of the report should be organised in the following sequence;

* Preliminary pages
* The body of the project(chapters)
* The bibliography
* Preliminary pages
* Approval Page
* The approval page is also known as signature page or completion certificate
* Internal as well as External guides should sign this page in order to assure that they have seen and approved the final version of the report.

4 Synopsis

* It should preferably be a single page (150 words max)
* The synopsis should be a summary or condensation of the project.

5 Acknowledgement

* The acknowledgment should not be more than one page.
* The student may acknowledge financial support, permission to use copy righted materials, trademarks, service marks, personal assistance etc.

6 Table of Contents

* It should be left justified, Times Roman 14 (student may use Table of content feature available in MS-WORD)

7 Introduction and Background

* It should have about 10% of total length
* Statement of Problem Area (brief, non-technical)
* Existing system, Methods and Procedures
* Background * Purpose/Objectives/justification of Project

8 Body of the Project Report

The body of the project report may include relevant features listed below:

* Company Profile
* Requirements Analysis
* Systems Design
* Database design (normalization, tables)
* File System and Data Structures
* User Interface design
* Prototyping

* Software and Hardware Platform (Selection of Operating System, Software packages, Computer Languages, Computer Systems and Peripherals)
* Verification and Software Testing
* DFD, Structure Charts, E-R diagram, Flowcharts, UML diagrams, Pseudocode, Decision Table, Decision Tree, Workflow, data dictionary
* Input and Output Forms/formats/reports
* Screen Dumps
* System Functional Specification
* Off-line or On-line Help feature
* Quality parameters/procedures
* Encryption/Security features
* Future Directions
* Results / conclusion
* References

Any points made in the text must be supported by evidence, either your results or the published findings of others. The sources are identified by citation.

9 To insure durability, permanency, and opacity, project report should be printed on A4 size white bond paper.

10 Typeface and Printing Chapter/Section Titles

* Should be printed in Times New Roman font in black colour
* Font size should be 16/14 points bold.
* Chapter should start on new page

11 Running text in the Report should be printed in Times New Roman font in black colour

* Font size should be 12 points.
* The print should be best quality
* Single line spacing for running text
* Double line spacing between paragraphs
* Printing on both sides

12 Margins

* Every page of the report, including all appendices, all notes, and the bibliography must have a LEFT and RIGHT margins of $1^{1/2}$ inches (to allow room for binding) and TOP and BOTTOM margins of 1 inch.
* Nothing should appear in LEFT and RIGHT margins. This means that all page numbers, text, tables, parts of illustrations, etc., must not appear in the margin area.

13 Page Numbers

* The page numbers must be bottom-cantered to the text (font Times New Roman; 10 points)
* All preliminary pages should NOT be numbered.
* The numbering should start from chapter-one (Introduction)
* Chapter titles (Headings) start on a new page.

* Leave an extra space after title
* Since you will have several levels of subheadings, distinguish one level from another in a consistent way, such as (1, 1.1, 1.2,2, 2.1,2.1.1,2.1.2, 2.2),
* Avoid having more than three levels of subheadings.

14 Length of Project Report

* In any case, the length of the graduate project report should not be more than 100 pages (excluding program listing)

15 Binding

* The project report must be clothbound (hardcover binding) and must be in light gray or black colour. Spiral binding is NOT allowed.

* The outside cover of the project report must follow the format described earlier and the lettering must be printed in gold letters.

Certificate for Evaluation

This is to certify that the undersigned have assessed and evaluated the Project Work titled "....." submitted by the student. The project Report has been (accepted / rejected) for the partial fulfilment of BCA programme.

Signature of the Examiner

Name of the Examiner

Sample Cover Page of the project report

JAIPUR NATIONAL UNIVERSITY

Jagatpura, Jaipur

info@jnujaipur. ac. in. *www.jnujaipujr. ac. in*

TITLE OF PROJECT REPORT

<Font Size 18><1.5 line spacing>

Submitted By

Student Name
Roll no

Under Guidance of Project Guide's Full Name

Submitted to the Jaipur National University, School of Computer and Systems Sciences in the partial fulfillment of the requirements for the

Award of Degree

Of

Bachelor of Computer Applications

MONTH & YEAR <Font Size 14

SCHOOL OF COMPUTER & SYSTEM SCIENCES